MASHEED

Master Product Specification (MPS)

Phase 1

Masheed ERP

Waqf by Masheed Investment Marketplace

Version: 1.0

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PART 1

Executive Summary

1.1 Overview

Masheed is a vertically integrated technology platform built specifically for the Saudi Arabian Awqaf sector.

Unlike generic ERP solutions that focus primarily on accounting or property management, Masheed is designed around the regulatory, operational, governance, and investment lifecycle of endowments.

The platform consists of two tightly integrated products:

1. Masheed ERP

An enterprise operating system for Awqaf organizations that enables them to manage every aspect of their operations, including:

- Waqf registration
- Governance
- Financial management
- Assets
- Beneficiaries
- Investments
- Documents
- Compliance
- AI-powered recommendations
- Expert consultations

The ERP is designed to ensure that every operational activity aligns with the conditions of the Waqif, supports regulatory compliance, and promotes long-term sustainability.

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2. Waqf by Masheed

A regulated digital investment marketplace that connects qualified Awqaf with investors.

The marketplace enables Awqaf organizations to publish investment opportunities after passing structured legal, financial, governance, and compliance reviews.

Investors can then evaluate, compare, and invest in opportunities through a transparent, data-rich platform.

1.2 Product Vision

To become the digital operating system for the Awqaf sector by providing the infrastructure required to manage, govern, grow, and invest in endowments while preserving the intentions of the Waqif and supporting national development goals.

1.3 Mission

To modernize the management of Awqaf through intelligent software that combines enterprise resource planning, governance, regulatory compliance, expert consultation, and investment enablement into a single integrated ecosystem.

1.4 Target Users

ERP

- Awqaf Organizations
- Private Awqaf
- Family Awqaf
- Public Awqaf
- Corporate Awqaf
- Foundations
- Nazers
- Board Members
- CEOs
- Financial Managers
- Asset Managers
- Auditors
- Compliance Officers
- Consultants
- Legal Advisors

Marketplace

- Individual Investors
- Institutional Investors
- Family Offices
- Foundations
- Investment Companies
- Endowment Organizations
- Banks (future)
- Real Estate Funds (future)

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2 Product Philosophy

Masheed is not an accounting system.

It is not a property management system.

It is not a CRM.

It is not an investment marketplace.

It is not a document management system.

Instead,

Masheed is an **Operating System for Awqaf**.

Every module exists to support one or more of the following objectives:

1. Protect the Waqf.
2. Preserve the Waqif's conditions.
3. Improve governance.
4. Ensure regulatory compliance.
5. Increase operational efficiency.
6. Enhance transparency.
7. Improve investment readiness.
8. Increase sustainability.
9. Enable informed decision-making through AI.
10. Maximize the long-term social and financial impact of the Waqf.

Every feature added to the platform must support at least one of these objectives.

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3 Core Design Principles

The entire platform should follow these principles.

Principle 1 — Waqif First

Every workflow, approval, financial transaction, investment, and distribution must respect the Waqif's documented conditions. The system should actively prevent actions that conflict with those conditions and provide clear explanations when a restriction is triggered.

Principle 2 — Compliance by Design

Compliance should be embedded into every workflow rather than treated as a separate checklist. Users should receive guidance and warnings before non-compliant actions are completed.

Principle 3 — Governance by Default

Governance should be part of the normal operation of the platform through structured approvals, audit logs, segregation of duties, conflict-of-interest declarations, and board oversight.

Principle 4 — AI as a Co-Pilot

Artificial intelligence should assist users by analyzing information, highlighting risks, recommending actions, and automating repetitive work. AI should never silently override human decisions or regulatory requirements.

Principle 5 — Transparency

Every important action within the system should be traceable, with complete visibility into who performed it, when it occurred, what changed, and why it changed.

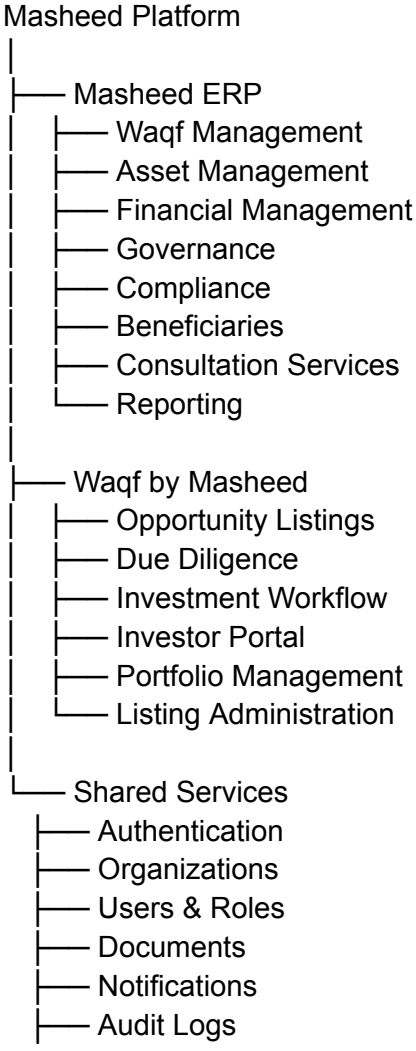
Principle 6 — One Source of Truth

Each Waqf should have a single, unified digital profile containing its legal documents, governance structure, assets, finances, beneficiaries, investments, compliance status, consultations, and historical records. This profile becomes the authoritative source for both ERP operations and investment marketplace eligibility.

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4 Product Ecosystem

The platform consists of two integrated products built on a shared foundation.



├── AI Services
└── Search

PART 2 — Masheed ERP

Chapter 1 — ERP Architecture

1.1 ERP Philosophy

Masheed ERP is not organized around accounting or departments. Instead, it is organized around the lifecycle of a Waqf.

Every piece of data belongs to a Waqf, and every action taken within the system contributes to one or more of the following objectives:

- Preserve the Waqif's conditions.
- Protect and grow Waqf assets.
- Ensure regulatory compliance.
- Improve governance.
- Support informed decision-making.
- Enable investment readiness.

This philosophy influences every module, workflow, permission, and AI recommendation throughout the platform.

1.2 High-Level ERP Structure

The ERP is divided into eight functional domains:

Masheed ERP

├── Foundation
│ ├── Organizations
│ ├── Users

- Roles & Permissions
- Settings
- Notifications
- Audit Logs

- Waqf Management
  - Waqf Profile
  - Waqif
  - Waqf Conditions
  - Beneficiaries
  - Governance
  - Compliance

- Asset Management
  - Properties
  - Land
  - Buildings
  - Equipment
  - Vehicles
  - Investments
  - Tokenization Readiness

- Finance
  - Accounting
  - Revenues
  - Expenses
  - Budgets
  - Distributions
  - Contracts
  - Financial Reports

- Operations
  - Tasks
  - Maintenance
  - Documents
  - Meetings
  - Approvals
  - Calendar

- Compliance
  - Regulations
  - Violations
  - Risk Register
  - Internal Audit

```
│   ├── AI Compliance Officer
│   └── Compliance Reports
│
├── Consultancy
│   ├── Experts
│   ├── Consultations
│   ├── Requests
│   ├── Recommendations
│   └── Knowledge Base
│
└── Marketplace Integration
    ├── Listing Readiness
    ├── Due Diligence
    ├── Listing Requests
    ├── Investment Documents
    └── Marketplace Sync
```

1.3 Core Platform Objects

Everything in Masheed revolves around a small number of core entities.

Organization

↓

Waqf

↓

Assets

↓

Income

↓

Expenses

↓

Beneficiaries

↓

Compliance

↓

Investment Readiness

↓

Marketplace

Every module references these core entities instead of maintaining isolated records.

1.4 Digital Twin Architecture

Each Waqf has a complete digital representation within the platform.

WAQF

Identity
│
├── Name
├── Type
├── Registration
├── Status
├── Documents
└── History

Governance
│
├── Board
├── Nazer
├── Policies
├── Committees
└── Meetings

Financial
│
├── Revenues

```
├── Expenses
├── Budget
├── Investments
├── Distributions
└── Statements

Assets
│
├── Properties
├── Land
├── Buildings
├── Equipment
├── Vehicles
└── Investments

Beneficiaries
│
├── Individuals
├── Organizations
├── Distribution Rules
└── Payment History

Compliance
│
├── Violations
├── Warnings
├── Risks
├── Audits
└── AI Monitoring

Marketplace
│
├── Readiness
├── Listings
├── Due Diligence
└── Investors

Consultations
│
├── Requests
├── Experts
├── Reports
└── Recommendations
```

This digital twin becomes the single source of truth for both the ERP and the investment marketplace.

1.5 Navigation Structure

The left sidebar should remain consistent across all pages.

Dashboard

Workspace

My Tasks

_____

WAQF

• Waqfs

• Waqifs

• Beneficiaries

• Governance

• Compliance

_____

Assets

• Properties

• Investments

• Maintenance

• Contracts

_____

Finance

• Accounting

• Revenues

• Expenses

• Budgets

• Distributions

Operations

• Calendar

• Documents

• Meetings

• Workflows

Consultancy

• Experts

• Requests

• AI Assistant

Marketplace

• Listing Readiness

• Listing Requests

• Marketplace Status

Reports

Analytics

Settings

1.6 Dashboard Philosophy

Every user should see a role-based dashboard.

There should never be a "generic dashboard."

Examples:

CEO Dashboard

- Total Waqf Value
- Asset Growth
- Revenue Trend
- Compliance Score
- Governance Score
- Investment Readiness
- High Risk Items
- AI Recommendations

Financial Manager

- Cash Position
- Outstanding Payments
- Budget Utilization
- Revenue
- Expenses
- Profitability
- Financial Risks

Nazer

- Compliance Alerts
- Beneficiary Issues
- Waqif Condition Violations
- Governance Tasks
- Pending Approvals
- AI Recommendations

Asset Manager

- Occupancy
- Maintenance
- Asset Health
- Valuation
- Insurance
- Rental Performance

Compliance Officer

- Violations
- Upcoming Deadlines
- Internal Audits
- Risk Register
- Missing Documents
- Conflict of Interest Declarations

1.7 Universal Object Layout

Every major object in the ERP should use a consistent layout.

For example:

Asset

Overview

Details

Financial

Documents

Activities

History

Compliance

AI Analysis

Relationships

The same pattern applies to:

- Waqf
- Property
- Beneficiary
- Contract
- Investment
- Consultation
- Listing
- Organization

This consistency reduces the learning curve for users.

1.8 Universal Record Header

Every record should display a standardized header with quick insights and actions.

Property #A-1024

Status

Compliance Score

Risk Score

Investment Score

AI Health Score

Owner

Created By

Last Updated

Marketplace Status

Quick Actions

Approve

Archive

Generate Report

Request Consultation

Open AI Assistant

View History

1.9 Global Search

Masheed should offer a platform-wide intelligent search.

Users can search by:

- Waqf Name
- Waqif
- Beneficiary
- Property
- Asset ID
- Contract
- Invoice
- Consultant
- Listing
- Investor
- Regulation
- AI Recommendation

Search results should include:

- Matching records
- Related records
- Recent activity
- AI-generated suggestions

Example:

Searching for "Al Noor Waqf" could return:

- Waqf Profile
- Associated Properties
- Financial Statements
- Active Compliance Issues
- Marketplace Listing
- Consultation History
- Related Documents

1.10 Universal Status System

Every workflow in the ERP should follow standardized lifecycle statuses.

Draft

↓

Submitted

↓

Under Review

↓

Pending Information

↓

Approved

↓

Active

↓

Suspended

↓

Completed

↓

Archived

Additional statuses can be added by specific modules, but these remain the foundation for consistency.

1.11 Cross-Module Relationships

One of Masheed's key differentiators is that modules are deeply interconnected rather than operating in isolation.

For example:

- A property is linked to its Waqf, revenue records, maintenance history, insurance policies, legal documents, and marketplace readiness.
- A beneficiary is connected to distribution rules, payment history, Waqif conditions, and audit records.
- A listing request references financial statements, governance documents, compliance status, and asset valuations automatically.

This interconnected model enables AI to analyze complete business contexts rather than fragmented data.

1.12 AI Everywhere

Every page in the ERP should include contextual AI assistance.

Instead of one generic chatbot, the AI understands the current screen and offers relevant actions.

Examples include:

- Explaining a regulation affecting the current Waqf.
- Detecting missing documents before submission.
- Summarizing financial performance.
- Suggesting governance improvements.
- Drafting consultation requests.
- Predicting maintenance needs.
- Highlighting marketplace readiness gaps.

The AI should always explain **why** it makes a recommendation and reference the supporting data, maintaining transparency and user trust.

PART 2 — ERP

Chapter 2 — Waqf Management Module

2.1 Purpose

The Waqf Management Module is the core of Masheed ERP.

Every other module within the platform is directly or indirectly connected to a Waqf.

Rather than treating a Waqf as a simple record, Masheed represents it as a complete digital entity that encapsulates its legal identity, governance structure, financial activities, assets, beneficiaries, regulatory status, historical changes, and investment readiness.

The module serves as the authoritative source of truth for all information related to an individual Waqf.

2.2 Objectives

The module should enable organizations to:

- Register and manage multiple Awqaf.
- Preserve and enforce the Waqif's intentions.
- Maintain complete legal documentation.
- Track governance structures.
- Monitor compliance.
- Link all assets and financial activities.
- Prepare Waqf for investment.
- Generate regulatory reports.
- Support AI-driven decision making.

2.3 Waqf Lifecycle

Every Waqf follows a standardized lifecycle.

Idea

↓

Draft

↓

Registration

↓

Legal Verification

↓

Governance Setup

↓

Asset Registration

↓

Financial Setup

↓

Operational

↓

Investment Ready

↓

Marketplace Listed

↓

Archived

Each stage unlocks additional capabilities while enforcing completion of mandatory requirements.

2.4 Waqf Profile Structure

Every Waqf profile consists of the following sections:

Overview

Legal Information

Founder (Waqif)

Waqf Conditions

Governance

Beneficiaries

Assets

Financial Overview

Compliance

Investment Readiness

Marketplace

Documents

Consultations

AI Insights

Timeline

Audit History

This unified structure allows users to navigate the complete context of a Waqf from a single location.

2.5 Waqf Header

Every Waqf page begins with a summary card.

WAQF NAME

Current Status

Registration Number

Organization

Total Asset Value

Annual Revenue

Beneficiaries

Compliance Score

Governance Score

Investment Readiness

Marketplace Status

AI Health Score

Quick Actions

Quick actions include:

- Edit
- Upload Document
- Add Asset
- Add Beneficiary
- Create Consultation
- Generate Report
- Request Marketplace Listing
- Open AI Assistant

2.6 Waqf Data Model

Section A — Identity

Field	Type	Required
Waqf ID	UUID	✓
Arabic Name	Text	✓
English Name	Text	
Short Name	Text	
Internal Code	Text	✓
Waqf Type	Dropdown	✓
Waqf Category	Dropdown	✓
Registration Number	Text	✓
Registration Date	Date	✓

Establishment Date	Date	
Status	Enum	✓
Description	Rich Text	
Vision	Rich Text	
Mission	Rich Text	

Section B — Classification

Fields

Public Waqf

Private Waqf

Family Waqf

Corporate Waqf

Educational

Health

Mosque

Real Estate

Agricultural

Cash Waqf

Investment Waqf

Mixed

Users may select multiple operational categories while maintaining a single legal classification.

Section C — Legal Information

Fields

Registration Authority

License Number

Commercial Registration

National Address

Tax Registration

Municipality Registration

Legal Representative

Legal Status

Expiration Dates

Official Website

Email

Phone

Government Portal Links

Section D — Address

Country

Region

City

District

Street

Building Number

Postal Code

Latitude

Longitude

Google Maps

Location Documents

2.7 Waqif (Founder) Information

Unlike traditional systems, Masheed treats the Waqif as a first-class entity with historical and legal significance.

Each Waqf may have one or more founders.

For each Waqif:

Field	Type
Waqif ID	UUID
Full Name	
Arabic Name	
Nationality	
National ID / Iqama	
Date of Birth	
Date of Death (if applicable)	
Contact Details	
Biography	
Relationship to Waqf	
Founder Type (Individual / Organization)	
Legal Documents	
Power of Attorney	
Notes	

The system maintains a permanent record even if the founder is deceased.

2.8 Waqf Conditions Engine (Core Differentiator)

This is one of Masheed's most important innovations.

Instead of storing the Waqf deed as a PDF, every condition is transformed into structured, machine-readable rules.

Each condition becomes an object with:

Field
Condition Number
Category
Original Arabic Text
Simplified Interpretation
AI Explanation
Related Assets
Related Beneficiaries
Related Investments
Related Financial Rules
Priority
Mandatory / Optional
Effective Date
Expiration Date
Current Status
Regulatory Notes

Each condition can then drive business logic throughout the platform.

For example:

- Restrict investments to certain asset classes.
- Limit distributions to specified beneficiaries.
- Require board approval for asset sales.
- Prevent use of funds outside defined purposes.

This engine ensures that operational decisions remain aligned with the Waqif's intent.

2.9 AI Waqif Interpreter

When a Waqf deed is uploaded, the AI should:

- Extract all conditions.
- Identify beneficiaries.
- Detect governing rules.
- Summarize the document.
- Highlight ambiguities.
- Suggest legal review if necessary.
- Generate structured conditions automatically.

The user reviews and confirms the extracted information before it becomes active.

2.10 Beneficiary Framework

Rather than simply listing beneficiaries, the system models their relationships and entitlements.

Each beneficiary record includes:

- Identity
- Type (Individual, Family, Organization, Public)
- Eligibility Criteria
- Supporting Documents
- Distribution Rules
- Payment Method
- Status
- Priority
- Historical Payments

- Related Conditions
- AI Eligibility Score

The framework supports future rule-based distribution automation while preserving manual oversight where required.

2.11 Waqf Timeline

Every important event becomes part of the Waqf's permanent history.

Examples include:

- Waqf created
- Deed updated
- Asset acquired
- Asset sold
- Beneficiary added
- Distribution completed
- Consultation requested
- Compliance issue detected
- Governance decision recorded
- Marketplace listing submitted
- Listing approved
- Listing closed

Users can filter the timeline by category, date, user, or event type.

2.12 Waqf Health Score (Masheed Innovation)

Every Waqf receives a continuously updated health score composed of multiple dimensions:

- Legal Completeness
- Governance Maturity
- Financial Stability
- Asset Quality
- Compliance Status

- Documentation Completeness
- Operational Performance
- Risk Profile
- Investment Readiness
- AI Confidence

Each dimension contributes to an overall score from 0 to 100, with recommendations for improvement.

This provides executives and regulators with a concise view of the Waqf's operational health.

2.13 Automations

The module should automatically:

- Detect missing legal documents.
- Notify users of expiring registrations.
- Validate required fields before activation.
- Generate onboarding tasks.
- Trigger governance setup after registration.
- Recommend consultations for detected issues.
- Calculate Health Score.
- Update Investment Readiness.
- Synchronize with the Marketplace module when eligibility changes.

2.14 Reports

The Waqf Management module should generate:

- Waqf Master Profile
- Legal Information Report
- Waqif Summary
- Beneficiary Register
- Conditions Register
- Governance Summary
- Compliance Overview
- Health Score Report
- Investment Readiness Report

- Executive Summary

All reports should be exportable in PDF and Excel, with configurable branding for each organization.

PART 2 — ERP

Chapter 3 — Asset Management

3.1 Purpose

The Asset Management Module enables Awqaf organizations to register, manage, monitor, maintain, value, develop, and optimize every asset owned by a Waqf.

The module serves as the operational foundation of the ERP because assets are the primary source of sustainability, revenue generation, and long-term impact.

Unlike conventional ERP systems, Masheed combines:

- Asset Registry
- Property Management
- Financial Performance
- Maintenance
- Compliance
- Governance
- Risk Analysis
- AI Intelligence
- Investment Readiness

into a unified asset management experience.

3.2 Objectives

The module shall:

- Maintain a complete digital inventory of assets.
- Protect Waqf assets.
- Maximize utilization.
- Improve maintenance planning.
- Monitor financial performance.
- Track legal ownership.
- Calculate asset value.
- Measure investment potential.
- Support Tokenization Readiness.
- Prepare assets for marketplace listing.

3.3 Asset Lifecycle

Every asset follows a structured lifecycle.

Identification

↓

Registration

↓

Legal Verification

↓

Valuation

↓

Operational

↓

Revenue Generation

↓

Maintenance

↓

Improvement

↓

Investment Evaluation

↓

Marketplace Ready

↓

Disposed (if legally permitted)

The system must record every transition with timestamps, approvals, and supporting documentation.

3.4 Supported Asset Types

Masheed should support a broad range of asset classes, including:

Real Estate

- Residential Buildings
- Commercial Buildings
- Towers
- Villas
- Hotels
- Apartments
- Offices
- Warehouses
- Retail Shops
- Mixed-use Developments

Land

- Agricultural Land

- Commercial Land
- Residential Land
- Industrial Land
- Undeveloped Land

Financial Assets

- Cash Waqf
- Investment Funds
- Sukuk
- Equities (where permitted)
- Murabaha Investments
- Deposits

Movable Assets

- Vehicles
- Equipment
- Machinery
- Furniture
- Technology Equipment

Intellectual Assets

- Trademarks
- Patents
- Copyrights
- Software
- Licenses

Other Assets

- Endowment Projects
- Operating Businesses
- Future Developments
- Renewable Energy Projects

3.5 Asset Object Structure

Each asset becomes a "Digital Asset Twin."

Asset

Overview

↓

Ownership

↓

Legal

↓

Financial

↓

Operations

↓

Maintenance

↓

Compliance

↓

Governance

↓

Risk

↓

Investment

↓

Marketplace

↓

AI Intelligence

↓

History

3.6 Asset Information

Identity

Field	Required
Asset ID	✓
Asset Code	✓
Arabic Name	✓
English Name	
Internal Reference	
Waqf	✓
Organization	✓
Asset Type	✓
Asset Category	✓
Asset Status	✓

Physical Information

Fields

GPS Coordinates

Address

District

City

Region

Country

Building Number

Parcel Number

Municipality Number

Land Area

Building Area

Number of Floors

Parking Spaces

Construction Year

Renovation Date

Occupancy Rate

Accessibility Rating

Energy Rating

Google Maps Link

Drone Images

360° Images

Virtual Tour

Blueprint Files

BIM Model (future)

Ownership

Fields

Ownership Type

Title Deed Number

Title Deed Copy

Government Registration

Registration Authority

Acquisition Date

Acquisition Method

Purchase Price

Current Owner

Previous Owner

Restrictions

Mortgage Status

Encumbrances

Legal Notes

Financial

Fields

Current Value

Book Value

Market Value

Replacement Cost

Insurance Value

Last Valuation

Valuation Company

Annual Revenue

Monthly Revenue

Expenses

Maintenance Cost

Net Income

ROI

Occupancy Revenue

Revenue Trend

Expense Trend

Projected Revenue

AI Revenue Forecast

Insurance

Fields

Provider

Policy Number

Coverage

Expiration

Premium

Claims

Risk Rating

Renewal Reminder

Compliance

Fields

Inspection Status

Legal Compliance

Environmental Compliance

Safety Compliance

Municipality Compliance

AI Compliance Score

Inspection Reports

Pending Violations

Deadlines

Governance

Fields

Approval Authority

Board Approval Required

Selling Restrictions

Leasing Restrictions

Waqif Restrictions

Committee

Approvals

Policy References

Marketplace

Fields

Investment Ready

Marketplace Eligible

Listing Status

Listing Date

Expected ROI

Investor Interest

Funding Status

Investment Score

3.7 Asset Dashboard

Every asset should display:

Asset Value

Monthly Revenue

Net Profit

Occupancy

Compliance

Maintenance

Risk

AI Health

Marketplace

Quick Actions

3.8 Maintenance Module

Every asset contains a full maintenance center.

Maintenance Types

- Preventive
- Corrective
- Emergency
- Predictive
- Inspection
- Cleaning
- Renovation

Every maintenance request includes:

- Request ID
- Priority
- Category
- Assigned Team
- Contractor
- Cost
- Start Date
- End Date
- Photos
- Documents
- Before Images
- After Images
- AI Summary
- Completion Report

3.9 AI Maintenance Planner

The AI continuously evaluates:

Age

Weather

Usage

Occupancy

Revenue

Previous Repairs

Inspection Results

Sensor Data (future)

and predicts:

Expected failure

Remaining useful life

Optimal maintenance date

Budget estimate

Recommended contractor type

Risk of postponement

3.10 Asset Risk Engine

Each asset receives a continuously updated risk assessment.

Risk Categories

Operational

Financial

Legal

Environmental

Insurance

Market

Governance

Compliance

Cyber (for smart assets)

Natural Disaster

Each risk includes:

Probability

Impact

Risk Owner

Mitigation Plan

Review Date

AI Recommendation

Status

3.11 AI Asset Intelligence

Instead of just reporting numbers, the AI should answer questions like:

"This property's rental yield is 18% below similar assets."

"Maintenance costs have increased by 22% over the last year."

"This building should undergo a new valuation."

"This asset qualifies for marketplace listing."

"This asset should not be listed because it conflicts with Waqif Condition #7."

"The insurance coverage expires in 38 days."

"This building has generated positive cash flow for 24 consecutive months."

3.12 Asset Health Score

Every asset receives a score from 0–100 based on weighted criteria:

Dimension	Weight
Legal Status	10%
Documentation Completeness	10%
Maintenance Condition	15%
Financial Performance	20%
Occupancy	10%
Compliance	15%
Governance	10%
Investment Readiness	10%

This score feeds into the overall Waqf Health Score.

3.13 Asset Relationships

Assets should never exist in isolation.

Each asset is linked to:

- Waqf
- Waqif Conditions
- Beneficiaries (where applicable)
- Revenue Records
- Expense Records
- Contracts
- Insurance Policies
- Maintenance Requests
- Legal Documents
- Valuations
- Risks
- Compliance Findings
- Consultations
- Marketplace Listings
- AI Recommendations

These relationships allow impact analysis. For example, if an asset's valuation drops significantly, the system can identify affected investment opportunities, revenue projections, and beneficiary distributions.

3.14 Smart Automations

The Asset Management module should automatically:

- Create recurring maintenance schedules.
- Alert users when insurance or legal documents are nearing expiration.
- Recommend asset revaluations based on market conditions or elapsed time.
- Detect abnormal expense spikes.
- Flag occupancy declines.
- Update investment readiness after major changes.
- Trigger consultation requests for high-risk assets.
- Synchronize asset information with the Marketplace module when listing eligibility changes.

3.15 Reports

The module should provide:

- Asset Register
- Asset Portfolio Summary
- Property Performance Report
- Occupancy Report
- Maintenance Report
- Insurance Report
- Valuation History
- Asset Health Report
- Risk Register
- Compliance Report
- Marketplace Eligibility Report
- Executive Portfolio Dashboard

Enhancement Proposal — "Asset Command Center"

Rather than treating assets as individual pages, I recommend creating an **Asset Command Center**.

This would be a visual control room where managers can:

- View all assets on an interactive map.
- Filter by Waqf, city, value, risk, occupancy, or compliance.
- See AI-generated alerts in real time.
- Compare asset performance across portfolios.
- Launch maintenance, inspections, or consultations directly from the map.
- Identify assets ready for investment listing with a single click.

This feature would be one of Masheed's strongest differentiators and give executives an immediate operational overview that is not commonly found in traditional ERP systems.

PART 2 — ERP
Chapter 4 — Financial Management
4.1 Purpose

The Financial Management module provides a complete financial operating system for Awqaf organizations. It combines accounting, budgeting, treasury, distributions, investment accounting, and financial governance into a unified platform that is aware of Waqf-specific regulations and the conditions of each Waqif.

Unlike conventional accounting systems, every financial transaction in Masheed is evaluated within the context of:

Waqf purpose.
Waqif conditions.
Beneficiary rights.
Governance policies.
Regulatory requirements.
Investment readiness.

The objective is not only to record financial activity but to ensure that all financial decisions are compliant, transparent, and aligned with the long-term sustainability of the Waqf.

4.2 Financial Architecture
Financial Management

├── Chart of Accounts
├── General Ledger
├── Journals
├── Cost Centers
├── Budgets
├── Revenues
├── Expenses
├── Treasury
├── Investments
├── Distributions
├── Financial Statements
├── Cash Flow
├── Forecasting
├── Compliance
├── AI Financial Intelligence
└── Marketplace Readiness

4.3 Chart of Accounts

Unlike generic ERPs, the chart of accounts should ship with a Waqf-optimized template, while allowing organizations to customize it.

Major Categories

Assets

Cash
Bank Accounts
Receivables
Investment Assets
Real Estate

Equipment
Other Assets

Liabilities

Payables
Accrued Expenses
Deposits
Financing (where legally applicable)

Equity / Endowment Funds

Principal Endowment
Restricted Endowment
Unrestricted Funds
Reserve Fund
Development Fund

Revenue

Rental Income
Agricultural Income
Investment Returns
Donations
Grants
Other Income

Expenses

Maintenance
Operations
Salaries
Utilities
Professional Services
Consultancy
Legal
Insurance
Administrative Costs

Beneficiary Distributions

Family Distribution
Educational Programs
Healthcare

Mosque Operations
Community Programs
Other Approved Uses

4.4 Financial Dimensions

Every transaction should support multiple dimensions.

Example:

Dimension	Example
Waqf	Al Noor Waqf
Asset	Tower A
Project	Expansion Project
Department	Operations
Cost Center	Maintenance
Beneficiary Group	Students
Funding Source	Rental Income
Condition Reference	Condition #5
Marketplace Impact	Yes

These dimensions allow advanced reporting and compliance validation.

4.5 Revenue Management

Supported revenue sources include:

Rental Income
Investment Returns
Donations
Cash Waqf Returns
Agricultural Revenue
Business Operations
Licensing
Service Revenue
Government Support
Other Approved Sources

Each revenue record includes:

Revenue ID
Revenue Type
Waqf
Asset
Customer/Tenant

Contract
Invoice
Payment Method
Amount
Tax Treatment
Currency
Collection Status
Collection Date
AI Risk Flag
Compliance Status

4.6 Expense Management

Every expense should be linked to its business purpose.

Required fields include:

Expense ID
Category
Vendor
Invoice
Cost Center
Asset
Waqf
Funding Source
Related Condition
Beneficiary Impact
Board Approval Required
Supporting Documents
Approval Workflow
AI Compliance Status

The system should prevent payment until required approvals are complete.

4.7 Treasury Management

Treasury features include:

Multi-bank account management.
Cash positions.
Scheduled payments.
Payment approvals.
Forecasted balances.
Investment liquidity.
Reserve monitoring.

Cash concentration reports.

The dashboard should show:

Available Cash
Restricted Funds
Upcoming Obligations
Expected Income
Liquidity Ratio
Burn Rate
AI Cash Forecast
4.8 Budget Management

Budget structure:

Annual Budget

↓

Departments

↓

Projects

↓

Assets

↓

Cost Centers

↓

Monthly Allocation

Budget capabilities:

Draft
Review
Approval
Revision
Freeze

Variance Analysis
Forecasting
4.9 Distribution Management

This module is unique to Awqaf.

Every beneficiary distribution must comply with:

Waqif Conditions
Distribution Rules
Beneficiary Eligibility
Available Funds
Governance Policies

Distribution workflow:

Generate Distribution Proposal

↓

AI Validation

↓

Condition Verification

↓

Beneficiary Verification

↓

Approval

↓

Payment

↓

Receipt

↓

Audit Log

Distribution fields:

Distribution ID
Beneficiary
Amount
Distribution Reason
Related Waqif Condition
Funding Source
Approval Reference
Payment Date
Payment Method
Status

4.10 Financial Compliance Engine

Before any transaction is posted, Masheed performs automated validation.

Checks include:

Is this expense allowed under the Waqif's conditions?
Is sufficient budget available?
Does the transaction require board approval?
Does it exceed approval limits?
Is the funding source valid?
Does it affect beneficiary allocations?
Are required documents attached?
Does it create a compliance violation?

Transactions that fail validation should be blocked or routed for additional review.

4.11 AI Financial Intelligence

The AI continuously analyzes financial data and provides insights such as:

Revenue growth trends.
Expense anomalies.
Budget overruns.
Cash flow risks.
Investment performance.
Distribution sustainability.
Forecasted liquidity.
Cost-saving opportunities.
Revenue optimization recommendations.

It should also explain why it generated each recommendation.

4.12 Marketplace Financial Readiness

The Marketplace depends on reliable financial information.

The ERP should calculate a Financial Readiness Score based on:

Availability of audited statements.
Revenue stability.
Cash flow health.
Asset profitability.
Debt exposure.
Budget discipline.
Documentation completeness.
Financial governance.

This score becomes part of the overall Listing Readiness assessment.

4.13 Financial Dashboard

The executive dashboard should include:

Executive KPIs
Total Assets
Net Asset Growth
Annual Revenue
Operating Expenses
Net Surplus
Distribution Ratio
Cash Position
Budget Utilization
Investment Return
Financial Health Score
AI Panel

Examples:

Rental income has increased by 12% compared to last year.

Maintenance costs for Tower A exceed the portfolio average by 18%.

Cash reserves will fall below the recommended threshold in approximately 95 days.

This Waqf is financially ready for marketplace listing.

4.14 Reports

The module should generate:

Financial Statements
Balance Sheet
Income Statement
Cash Flow Statement
Trial Balance
General Ledger
Journal Entries
Waqf Reports
Revenue Analysis
Expense Analysis
Distribution Report
Budget Variance
Treasury Report
Financial Health Report
Marketplace Readiness
Funding Source Report
Executive Reports
Financial Performance Dashboard
AI Financial Summary
Forecast Report
Investment Return Analysis
Sustainability Report
4.15 Financial Health Score (Masheed Innovation)

Each Waqf receives a Financial Health Score from 0–100 based on weighted dimensions:

Dimension	Weight
Liquidity	15%
Revenue Stability	20%
Expense Control	15%
Budget Adherence	10%
Distribution Sustainability	15%
Asset Profitability	15%
Documentation Completeness	5%
Compliance	5%

The score contributes to the overall Waqf Health Score and is displayed on executive dashboards.

4.16 Approval Matrix (Enterprise Feature)

Financial approvals should be fully configurable.

Example:

Transaction Type	Amount	Required Approvals
Expense	≤ SAR 10,000	Department Manager
Expense	SAR 10,001–100,000	Manager + Finance
Expense	> SAR 100,000	Finance + CEO + Nazer
Asset Sale	Any Amount	Board + Nazer + Legal
Beneficiary Distribution	Configurable	Based on Waqf policy

Approval limits should be configurable per organization and per Waqf.

4.17 Audit & Traceability

Every financial action must be fully traceable.

For each transaction, the system records:

Who created it.
Who modified it.
Who approved it.
Approval timestamps.
Linked documents.
Related Waqif conditions.
AI validation results.
Compliance checks.
Before/after changes.
Reason for changes.

Nothing should be permanently deleted; records should be versioned and auditable.

Enhancement Proposal — "Financial Command Center"

Instead of a traditional accounting dashboard, Masheed should include a Financial Command Center that combines accounting, compliance, governance, and AI into one executive view.

It should answer questions such as:

Which Waqf requires immediate financial attention?
Which assets generate the highest returns?
Which distributions are pending approval?
Which expenses may violate Waqif conditions?
Which Waqf is financially ready to be listed on Waqf by Masheed?
Which financial risks could affect long-term sustainability?

This transforms finance from a historical reporting function into a proactive decision-support system.

PART 2 — ERP

Chapter 5 — Governance & Compliance Engine

5.1 Philosophy

Governance should never be something users think about once a year.

It should happen automatically.

Every approval.

Every meeting.

Every payment.

Every investment.

Every beneficiary.

Every policy.

Every decision.

Everything should automatically follow governance rules.

The user should never ask

> "Am I compliant?"

The system should already know.

5.2 Governance Architecture

Governance Engine

```
├── Board Management
├── Nazer Management
├── Committees
├── Decision Register
├── Policies
├── Conflict of Interest
├── Delegation Matrix
├── Internal Controls
├── Risk Management
├── Compliance Engine
├── Internal Audit
├── Regulatory Calendar
├── AI Governance Advisor
└── Governance Reports
```

5.3 Governance Dashboard

The dashboard should immediately answer:

Governance Health

95%

Compliance Score

91%

Board Meetings

4 Pending

Policies

36 Active

2 Need Review

Internal Audit

3 Findings

Critical Risks

5

AI Recommendations

8

Marketplace Readiness

88%

5.4 Governance Score

Every Waqf receives a Governance Score.

The score is calculated using:

Component	Weight
Board Governance	15%
Policies	10%
Risk Management	15%
Internal Controls	10%
Compliance	20%
Transparency	10%
Documentation	10%
Internal Audit	10%

The Governance Score becomes one of the primary KPIs of the platform.

5.5 Board Management

Every Waqf should maintain a structured digital board.

Each member includes:

- Name
- Role
- Appointment Date
- Term
- Qualifications
- Contact Information
- Independence Status
- Committees
- Voting Rights

- Attendance
- Conflict of Interest Declaration
- Documents
- Performance Review
- Status

Board Dashboard

- Members
- Attendance
- Decisions
- Voting History
- Committees
- Upcoming Meetings
- Expiring Terms

5.6 Committee Management

Supported committees:

- Investment
- Audit
- Governance
- Risk
- Finance
- Asset Management

Organizations can create additional committees.

Each committee includes:

Purpose

Members

Responsibilities

Authority

Meeting Schedule

Decisions

Performance

5.7 Meeting Management

Meeting workflow

Create Meeting

↓

Agenda

↓

Invite Members

↓

AI Agenda Suggestions

↓

Meeting

↓

AI Minutes

↓

Voting

↓

Decision Register

↓

Tasks

↓

Follow-up

Meeting fields

Meeting ID

Meeting Type

Committee

Location

Virtual Link

Agenda

Attendees

Quorum

Minutes

Attachments

Voting Results

Decisions

Tasks

Recording

AI Summary

5.8 Decision Register

One of Masheed's strongest differentiators.

Every decision becomes searchable forever.

Fields

Decision ID

Meeting

Decision Type

Description

Reason

Related Assets

Related Waqf

Related Investment

Related Beneficiaries

Approvals

Responsible Person

Implementation Date

Status

AI Impact Analysis

Example

Decision

Sell Property

↓

AI

"This decision affects:

• Annual revenue

• Beneficiary distributions

• Marketplace listing

• Waqif Condition #4

Please review."

5.9 Policy Management

Organizations should manage every internal policy inside Masheed.

Examples

Investment Policy

Conflict of Interest Policy

Procurement Policy

Asset Policy

Maintenance Policy

Risk Policy

Financial Policy

Consultation Policy

Document Policy

Marketplace Policy

Every policy includes

Version

Approval Date

Review Date

Owner

Documents

Related Regulations

Related Workflows

Training Status

AI Summary

5.10 Conflict of Interest Module

Every executive

Every board member

Every consultant

Every approver

must periodically submit a declaration.

Fields

Declaration Date

Organization

Relationships

Financial Interests

Related Companies

Related Assets

Family Relationships

Potential Conflicts

Supporting Documents

Approval

AI Conflict Analysis

AI example

"Board Member A owns Company X."

↓

Company X submitted quotation.

↓

AI blocks approval.

↓

Governance Alert.

5.11 Delegation Matrix

Every approval follows configurable authority.

Example

Expense

↓

Department Head

↓

Finance

↓

CEO

↓

Nazer

↓

Board

Approval authority depends on:

Amount

Asset

Risk

Waqf

Investment

Policy

Users never need to memorize approval rules.

The system routes them automatically.

5.12 Risk Register

Every risk is managed centrally.

Risk Categories

Financial

Operational

Legal

Compliance

Investment

Cybersecurity

Asset

Reputation

Governance

Strategic

Market

Environmental

Every Risk

Risk ID

Category

Description

Probability

Impact

Owner

Controls

Mitigation

Target Date

Residual Risk

AI Risk Score

AI predicts

Emerging risks

Trend

Mitigation

Priority

5.13 Internal Controls Library

Every process can include controls.

Example

Payment

↓

Invoice Required

↓

Approval Required

↓

Budget Check

↓

Condition Check

↓

Compliance Check

↓

Payment

Controls are configurable.

5.14 Internal Audit

Audit Types

Financial

Asset

Governance

Compliance

Investment

IT

Operational

Audit workflow

Planning

↓

Checklist

↓

Evidence

↓

Findings

↓

Recommendations

↓

Management Response

↓

Follow-up

↓

Closed

5.15 Regulatory Calendar

Every regulatory deadline appears automatically.

Examples

Annual Reports

Financial Statements

Board Reviews

Policy Reviews

Asset Valuation

Insurance Renewal

Registration Renewal

Marketplace Renewal

Consultation Review

Audit Schedule

AI sends reminders before deadlines.

5.16 Compliance Engine

This is where Masheed becomes truly unique.

Instead of storing regulations,

Masheed stores **Compliance Rules**.

Example

Rule

Annual Asset Valuation

↓

Frequency

12 months

↓

Applies To

Real Estate

↓

Responsible

Asset Manager

↓

Deadline

365 Days

↓

Penalty

Medium

↓

Automation

Generate Task

Notify Manager

Update Dashboard

If overdue

↓

Decrease Compliance Score

Every regulation becomes executable logic.

5.17 Compliance Dashboard

Shows

Compliance %

Violations

Warnings

Upcoming Deadlines

Pending Reviews

Risk Level

Marketplace Eligibility

AI Summary

Trend

5.18 Violation Management

Every violation includes:

Violation ID

Related Regulation

Category

Description

Severity

Date

Detected By

Assigned To

Root Cause

Corrective Action

Evidence

Due Date

Status

AI Recommendation

Lessons Learned

Repeat Count

This creates a historical compliance record for each Waqf.

5.19 AI Governance Advisor

Unlike a chatbot, this is a proactive governance assistant.

It continuously monitors:

- Governance documents.
- Board decisions.
- Policy reviews.
- Audit findings.
- Risks.
- Financial approvals.
- Waqif conditions.
- Marketplace readiness.

Example recommendations:

> "The Investment Policy is due for review in 15 days."
>
> "Two board members have missed three consecutive meetings."
>
> "An upcoming asset sale requires board approval under your delegation matrix."
>
> "This marketplace listing cannot proceed until the latest valuation is uploaded."

Every recommendation includes the underlying reasoning and links directly to the affected records.

5.20 Governance Reports

The module should generate:

Governance Reports

- Governance Health Report
- Board Effectiveness Report
- Committee Performance
- Decision Register
- Policy Status Report
- Delegation Matrix Report

Compliance Reports

- Compliance Scorecard
- Violation Register
- Regulatory Calendar
- Compliance Trend Analysis
- Corrective Action Status

Risk Reports

- Enterprise Risk Register
- Heat Map
- Top 10 Risks
- Risk Trend
- Mitigation Progress

Audit Reports

- Annual Audit Plan

- Audit Findings
- Management Responses
- Outstanding Actions
- Audit Closure Report

5.21 Executive Command Center (Masheed Flagship Feature)

This is the screen I want every CEO and Nazer to open every morning.

It combines:

Organization Health

↓

Waqf Health

↓

Governance

↓

Compliance

↓

Financial

↓

Assets

↓

Marketplace

↓

AI Priorities

↓

Critical Actions Today

Instead of searching through modules, executives immediately see:

- The three most urgent compliance issues.
- Waqf Health Score by portfolio.
- High-risk assets.
- Pending board decisions.
- Marketplace opportunities blocked by missing governance items.
- AI-recommended actions ranked by impact.

5.22 Governance Integration with Waqf by Masheed

Governance should directly influence marketplace eligibility.

A listing **cannot** proceed if required governance conditions are not met.

The Marketplace receives governance signals such as:

- Governance Score.
- Compliance Score.
- Outstanding violations.
- Required approvals.
- Current audit status.
- Asset valuation validity.
- Conflict-of-interest status.
- Board approval confirmation.
- Policy compliance.

This ensures only qualified and well-governed opportunities are presented to investors.

Major Architectural Recommendation (Most Important So Far)

I believe **Masheed needs one more foundational engine** before we build the Marketplace:

The Workflow Engine

Almost every enterprise platform (Salesforce, ServiceNow, SAP, Microsoft Dynamics) has a configurable workflow engine.

For Masheed, **every process should be workflow-driven**, not hard-coded.

Examples:

- New Waqf registration.
- Asset acquisition.
- Asset sale.
- Beneficiary approval.
- Consultation request.
- Marketplace listing.
- Financial approval.
- Policy review.
- Risk mitigation.
- Audit findings.
- Board decisions.

Each workflow should be configurable with:

- Stages.
- Required fields.
- Required documents.
- Approval rules.
- Notifications.
- SLA timers.
- AI checkpoints.
- Escalation rules.
- Completion criteria.

PART 3 — Waqf by Masheed

Chapter 1 — Product Architecture

1.1 Vision

Waqf by Masheed is a regulated investment marketplace that enables qualified Awqaf organizations to present investment opportunities to investors through a transparent, governed, and AI-assisted platform.

Unlike traditional crowdfunding or investment platforms, every opportunity originates from Masheed ERP, where its governance, compliance, financial health, asset information, and legal documentation are continuously synchronized.

This ensures investors always evaluate opportunities using current and verified information rather than static documents.

1.2 Objectives

The marketplace should:

- Increase access to capital for Awqaf.
- Improve investment transparency.
- Standardize due diligence.
- Reduce manual paperwork.
- Increase investor confidence.
- Ensure regulatory compliance.
- Connect investors with qualified opportunities.
- Monitor opportunities throughout their lifecycle.
- Enable continuous reporting after investment.

1.3 Marketplace Philosophy

Investment should never begin with uploading a PDF.

Investment begins with trust.

Trust is generated through verified operational data from the ERP.

Every listing should answer four key questions:

- Is the Waqf well governed?

- Is the opportunity financially sound?
- Does the opportunity comply with the Waqif's conditions?
- Can investors make informed decisions based on reliable data?

1.4 Platform Architecture

Waqf by Masheed

├── Marketplace Home
├── Opportunity Explorer
├── Investment Opportunities
├── Investor Dashboard
├── Waqf Profiles
├── Due Diligence Center
├── AI Investment Assistant
├── Portfolio
├── Reporting Center
├── Notifications
├── Documents
└── Administration

1.5 User Types

Investors

- Individual Investors
- Family Offices
- Companies
- Foundations
- Institutional Investors
- Government Investors (future)

Opportunity Owners

- Waqf Organizations

- Nazer
- CEOs
- Investment Managers

Platform Administrators

- Listing Review Team
- Compliance Officers
- Legal Review
- Financial Analysts
- Governance Review
- Marketplace Managers

Consultants

- Legal Advisors
- Financial Consultants
- Valuation Experts
- Engineering Consultants
- Sharia Advisors
- Governance Specialists

Unlike the ERP, consultants can be invited into the marketplace to participate in due diligence without accessing unrelated organizational data.

1.6 Marketplace Navigation

Home

Discover

Investment Opportunities

Saved Opportunities

My Investments

Reports

Messages

Documents

Notifications

AI Investment Advisor

Profile

Settings

1.7 Opportunity Lifecycle

Every opportunity follows a structured lifecycle.

Draft

↓

Internal Review

↓

Document Collection

↓

Financial Review

↓

Legal Review

↓

Governance Review

↓

Compliance Review

↓

Valuation Verification

↓

Risk Assessment

↓

Investment Committee

↓

Approved

↓

Published

↓

Open for Investment

↓

Fully Funded

↓

Operational

↓

Performance Monitoring

↓

Closed

Unlike most marketplaces, opportunities remain active after funding to provide ongoing reporting and oversight.

1.8 Opportunity Profile

Each investment opportunity should contain:

Overview

↓

Investment Summary

↓

Waqf Profile

↓

Asset Details

↓

Financial Analysis

↓

Expected Returns

↓

Timeline

↓

Governance

↓

Compliance

↓

Risk Analysis

↓

Documents

↓

Updates

↓

Questions & Answers

↓

Investment Terms

↓

Performance Reports

This structure gives investors a comprehensive view before committing funds.

1.9 Opportunity Header

Every listing displays a standardized summary.

Opportunity Name

Investment Type

Funding Goal

Minimum Investment

Target Return

Funding Progress

Investment Period

Governance Score

Compliance Score

Risk Rating

Marketplace Rating

Status

Invest Button

Save

Share

1.10 Opportunity Information

Identity

- Opportunity ID
- Opportunity Name (Arabic & English)
- Opportunity Code
- Related Waqf
- Related Asset(s)
- Opportunity Type
- Sector
- Description
- Objectives

Financial

- Funding Goal
- Minimum Investment
- Maximum Investment (if applicable)
- Funding Model
- Investment Currency
- Project Cost
- Expected Revenue
- Expected Expenses

- Net Return Projection
- Break-even Point
- Exit Strategy (if applicable)

Timeline

- Submission Date
- Approval Date
- Funding Start
- Funding End
- Project Start
- Estimated Completion
- Reporting Frequency

Governance

Automatically synchronized from the ERP:

- Governance Score
- Board Approval Status
- Policy Compliance
- Active Audit Findings
- Delegation Approval
- Governance Notes

Compliance

Automatically synchronized:

- Compliance Score
- Outstanding Violations
- Regulatory Status
- Required Licenses
- Legal Restrictions
- Waqif Condition Validation

Risk

The marketplace should present a structured risk profile covering:

- Market Risk
- Financial Risk
- Operational Risk
- Construction Risk (where applicable)
- Legal Risk
- Governance Risk
- Compliance Risk
- Environmental Risk

Each risk includes:

- Severity
- Likelihood
- Mitigation Actions
- Current Status

1.11 Investor Dashboard

Each investor should have a personalized dashboard showing:

- Active Investments
- Portfolio Value
- Expected Returns
- Received Returns
- Investment Performance
- Pending Opportunities
- Saved Opportunities
- Watchlist
- AI Recommendations
- Notifications
- Required Actions

1.12 AI Investment Advisor

Rather than simply recommending opportunities, the AI should provide explainable investment insights.

Examples:

- "This opportunity has demonstrated stable rental income over the past three years."
- "Funding has reached 72%, with participation from three institutional investors."
- "The project's compliance score improved after completing the latest governance review."
- "Projected maintenance costs may affect expected returns in Year 4."

The AI should also answer investor questions using verified ERP data rather than generating unsupported assumptions.

1.13 Marketplace KPIs

The marketplace should monitor:

Platform Metrics

- Total Listed Opportunities
- Active Opportunities
- Total Capital Raised
- Average Funding Time
- Investor Growth
- Funding Success Rate
- Average Governance Score
- Average Compliance Score

Opportunity Metrics

- Page Views
- Investor Interest
- Saved Opportunities
- Questions Submitted
- Conversion Rate
- Funding Progress
- Time to Funding

These metrics help administrators optimize the platform and identify high-performing opportunity types.

1.14 Marketplace Notifications

Events that trigger notifications include:

- Listing submitted.
- Review started.
- Additional information requested.
- Governance issue detected.
- Listing approved.
- Listing published.
- New investor question.
- Investment received.
- Funding milestone reached.
- Opportunity fully funded.
- Periodic performance report published.
- Compliance issue affecting an active listing.

1.15 Integration with Masheed ERP

The marketplace should **never duplicate ERP data**.

Instead, it consumes approved information from the ERP through controlled synchronization.

Examples:

Pulled from ERP

- Waqf Profile
- Asset Details
- Financial Summaries
- Governance Score
- Compliance Score
- Health Score
- Documents
- AI Readiness Score

Managed within Marketplace

- Funding Progress
- Investor Activity
- Questions & Answers

- Marketing Content
- Public Updates
- Investor Communications

This architecture ensures consistency and eliminates conflicting records.

Architectural Recommendation — Opportunity Digital Twin

Just as every Waqf has a Digital Twin in the ERP, every marketplace opportunity should have its own **Opportunity Digital Twin**.

It should continuously aggregate:

- Live ERP data.
- Funding status.
- Investor engagement.
- Project milestones.
- Financial performance.
- Compliance updates.
- Governance changes.
- AI insights.

This means an opportunity is no longer a static listing but a living investment record that evolves throughout its lifecycle. It also lays the foundation for future features such as secondary investment markets, performance benchmarking, and advanced portfolio analytics without redesigning the platform architecture.

PART 3 — Waqf by Masheed

Chapter 2 — Listing Management & Due Diligence Engine

2.1 Philosophy

A listing is not a document.

A listing is a certification that a Waqf has reached a level of operational, financial, governance, and regulatory maturity that makes it suitable for investment.

The Marketplace should therefore reject the idea of manually reviewing random submissions.

Instead, Masheed continuously evaluates readiness using data already available in the ERP.

Listing becomes the final step—not the first.

2.2 Listing Sources

An opportunity can originate from three sources:

A. From Masheed ERP (Recommended)

The organization is already using Masheed ERP.

The system automatically identifies assets or projects that are investment-ready and suggests creating a listing.

B. Direct Marketplace Submission

Organizations not yet using the ERP can submit a listing manually.

The system creates a temporary workspace where they complete all required information before review.

This onboarding flow should encourage migration to the full ERP over time.

C. Administrator-Initiated

Marketplace administrators may create listings on behalf of organizations participating in strategic programs or government initiatives.

2.3 Listing Lifecycle

Draft

↓

Readiness Assessment

↓

Document Collection

↓

Financial Review

↓

Legal Review

↓

Governance Review

↓

Compliance Review

↓

Technical Review

↓

Risk Review

↓

Investment Committee

↓

Approved

↓

Published

↓

Funding

↓

Monitoring

↓

Completed

Each stage has:

- Required fields
- Required documents
- Responsible users
- SLA
- AI validation
- Approval requirements
- Exit criteria

2.4 Listing Workspace

Every listing should have its own workspace.

Overview

↓

Progress

↓

Documents

↓

Financials

↓

Legal

↓

Governance

↓

Compliance

↓

Technical

↓

Risks

↓

AI Analysis

↓

Questions

↓

Review History

↓

Approvals

Users never leave the workspace during the listing process.

2.5 Listing Readiness Score

Instead of saying

Ready / Not Ready

Masheed calculates a **Listing Readiness Score (0–100).**

Category	Weight
Asset Quality	15%
Financial Readiness	20%
Governance	15%
Compliance	20%
Documentation	10%
Risk Profile	10%
Market Attractiveness	10%

Example

Overall Readiness

89%

Financial

94%

Governance

92%

Compliance

97%

Documentation

80%

Risk

85%

Recommendation

Upload the latest engineering report
before submitting.

2.6 Required Information

Every listing should collect structured information.

Opportunity Information

- Opportunity Name
- Opportunity Code
- Opportunity Type
- Investment Category
- Sector
- Description
- Objectives
- Expected Impact
- Funding Model
- Funding Goal
- Investment Period
- Expected Exit Strategy
- Currency

Related Waqf

- Waqf
- Organization
- Related Asset(s)
- Related Project(s)
- Investment Manager
- Nazer
- Primary Contact

Financial Information

- Capital Requirement
- Project Budget
- Revenue Projection
- Expense Projection
- Cash Flow Projection
- IRR
- ROI
- NPV
- Payback Period
- Funding Sources
- Financial Assumptions

Legal Information

- Ownership Documents
- Title Deeds
- Government Approvals
- Licenses
- Contracts
- Easements
- Existing Obligations
- Pending Litigation
- Legal Opinion

Governance

Automatically synchronized:

- Governance Score
- Board Approval
- Committee Approval
- Policy Compliance
- Conflict of Interest Status
- Delegation Approval

Compliance

Automatically synchronized:

- Compliance Score
- Outstanding Violations
- Regulatory Status
- Required Reports
- Valuation Status
- Waqif Condition Validation

Technical Information

- Engineering Reports
- Site Plans
- Images
- Videos
- BIM Models (future)
- Feasibility Study
- Environmental Studies
- Market Study

2.7 Required Documents

The system should maintain a configurable checklist.

Examples include:

Legal

- Waqf Deed
- Registration Certificate
- Title Deeds
- Commercial Registration (if applicable)
- Government Licenses

Financial

- Audited Financial Statements
- Budget
- Cash Flow
- Valuation Report
- Revenue History

Governance

- Board Resolution
- Committee Approval
- Investment Policy
- Governance Report

Technical

- Engineering Report
- Feasibility Study
- Architectural Drawings
- Environmental Assessment
- Market Analysis

Administrators should be able to customize required documents by opportunity type.

2.8 AI Document Review

The AI should review every uploaded document.

Examples:

- Missing signatures.
- Expired documents.
- Inconsistent values.
- Missing pages.
- Low scan quality.

- Incorrect document type.
- Conflicting information.
- Missing approvals.

The AI provides a confidence score and flags items requiring human review.

2.9 Due Diligence Engine

Instead of a single review step, Masheed divides due diligence into specialized tracks.

Financial

↓

Legal

↓

Technical

↓

Governance

↓

Compliance

↓

Risk

↓

Investment

↓

Final Review

Each track can progress independently, reducing bottlenecks.

2.10 Review Assignments

Each review is assigned to one or more reviewers.

Reviewer fields:

- Reviewer
- Role
- Department
- Assigned Date
- Due Date
- Status
- Comments
- Recommendation
- Attachments
- AI Summary

The system tracks turnaround time and SLA compliance.

2.11 Issue Management

Reviewers create structured issues rather than exchanging emails.

Issue fields:

- Issue ID
- Category
- Severity
- Description
- Requested Action
- Assigned To
- Due Date
- Resolution
- Evidence
- Status

Issues remain linked to the listing for future audits.

2.12 AI Listing Advisor

The AI continuously evaluates the listing.

Examples:

> "The projected occupancy rate is significantly higher than comparable projects."
>
> "The valuation report expires in 18 days."
>
> "This opportunity meets governance requirements but is missing an environmental assessment."
>
> "Funding assumptions rely heavily on optimistic rental growth."

Each recommendation links to supporting data and relevant regulations where applicable.

2.13 Listing Approval Matrix

Every listing follows configurable approvals.

Example:

Stage	Required Approver
Financial Review	Financial Analyst
Legal Review	Legal Advisor
Governance Review	Governance Officer
Compliance Review	Compliance Officer
Technical Review	Engineering Reviewer
Investment Review	Investment Committee
Final Approval	Marketplace Director

Organizations can configure additional approval steps based on investment size or asset class.

2.14 Public Listing Generation

After approval, Masheed automatically generates the public-facing investment page.

Public information includes:

- Opportunity Summary
- Investment Highlights
- Funding Goal
- Timeline
- Expected Returns
- Asset Overview
- Governance & Compliance Badges
- Key Documents approved for public release
- FAQs
- Contact Information

Sensitive ERP data remains private.

2.15 Post-Publication Monitoring

Publication is not the end of the workflow.

The platform continuously monitors:

- Governance changes.
- Compliance violations.
- Financial deterioration.
- Asset status.
- Major legal events.
- Expired documents.
- Risk level changes.

If critical thresholds are crossed, the platform can:

- Display investor warnings.
- Suspend new investments.

- Request corrective actions.
- Escalate to administrators.

2.16 Listing Analytics

Each listing should display analytics such as:

- Views
- Unique Visitors
- Investor Interest
- Saved Listings
- Questions Submitted
- Average Time on Page
- Conversion Rate
- Funding Progress
- Funding Velocity
- Completion Rate

These insights help organizations improve future opportunities.

Major Recommendation — Investment Data Room

I strongly recommend that every listing automatically includes a secure **Investment Data Room**, similar to those used in private equity and institutional transactions.

Instead of attaching documents to emails, all approved documents are stored in a permission-controlled workspace with:

- Folder-based organization.
- Version history.
- Watermarked downloads.
- Granular access permissions.
- Activity logs (who viewed or downloaded what).
- Q&A linked to specific documents.
- Digital acknowledgments (e.g., confidentiality acceptance).
- AI-powered document search and summarization.

This elevates Waqf by Masheed from a simple marketplace to an institutional-grade investment platform suitable for large investors.

PART 3 — Waqf by Masheed

Chapter 3 — Investor Management & Portfolio Intelligence

3.1 Purpose

The Investor Management module provides a comprehensive digital experience for individuals and institutions investing through Waqf by Masheed.

It enables investors to discover opportunities, complete onboarding, manage investments, monitor portfolio performance, receive reports, participate in governance communications where applicable, and receive AI-powered insights throughout the investment lifecycle.

Rather than focusing only on transactions, the platform aims to build long-term investor confidence through transparency, continuous reporting, and verified operational data synchronized from Masheed ERP.

3.2 Investor Lifecycle

Every investor follows a structured journey.

Visitor

↓

Register

↓

Identity Verification

↓

Profile Completion

↓

Investment Preferences

↓

Risk Profile

↓

Eligible Investor

↓

Investment

↓

Portfolio Management

↓

Performance Monitoring

↓

Reinvestment

The system should preserve the complete history of the investor relationship.

3.3 Investor Types

Supported investor categories include:

Individual

- Saudi Citizens
- GCC Citizens
- Residents (where permitted)
- International Investors (future)

Organizations

- Companies
- Family Offices
- Foundations
- Charitable Organizations
- Financial Institutions
- Government Entities

Classification

Investor profiles should include configurable classifications such as:

- Retail
- Professional
- Institutional
- Strategic Partner
- Anchor Investor

These classifications may influence investment limits, approval requirements, and available opportunities.

3.4 Investor Profile

Each investor has a unified profile containing:

Identity

- Investor ID
- Investor Type
- Arabic Name
- English Name

- Nationality
- National ID / Commercial Registration
- Date of Registration
- Status

Contact

- Mobile
- Email
- Address
- Preferred Language
- Preferred Communication Method

Investment Information

- Preferred Investment Size
- Preferred Sectors
- Geographic Preferences
- Investment Objectives
- Investment Experience
- Investment Horizon
- Liquidity Preference

Compliance

- Identity Verification Status
- Document Status
- Risk Classification
- Approval Status
- Internal Notes

Activity

- Opportunities Viewed

- Saved Opportunities
- Active Investments
- Historical Investments
- Messages
- Documents
- Notifications

3.5 Investor Dashboard

Every investor should have a personalized dashboard.

Portfolio Value

↓

Active Investments

↓

Projected Returns

↓

Received Returns

↓

Investment Performance

↓

Watchlist

↓

New Opportunities

↓

AI Recommendations

↓

Pending Actions

Dashboard Widgets

Portfolio Summary

Total Invested

Current Portfolio Value

Projected Return

Portfolio Growth

Cash Available

Upcoming Distributions

Portfolio Risk

Average Governance Score

Average Compliance Score

3.6 Portfolio Management

Instead of showing a list of investments, Masheed builds a living portfolio.

Each investment displays:

- Investment Amount
- Ownership Percentage (where applicable)
- Investment Date
- Opportunity Status
- Current Performance
- Project Milestones
- Expected Returns
- Received Returns
- Risk Rating
- Governance Status

- Compliance Status

Portfolio filters

By Waqf

By Asset

By Sector

By Return

By Risk

By Status

By Investment Year

3.7 Portfolio Analytics

The platform should calculate:

Total Portfolio Value

Total Invested Capital

Projected Annual Return

Historical Return

Portfolio Diversification

Average Risk

Average Governance Score

Average Compliance Score

Cash Yield

Asset Allocation

Geographic Allocation

Sector Allocation

Investment Concentration

3.8 Investment Timeline

Every investment includes a timeline.

Opportunity Published

↓

Investment Submitted

↓

Investment Approved

↓

Funding Completed

↓

Project Started

↓

Milestone Updates

↓

Revenue Generation

↓

Returns Distributed

↓

Investment Closed

Every milestone should contain supporting documents, updates, and communications.

3.9 Returns Management

The system should track:

Projected Returns

Actual Returns

Distribution Schedule

Payment History

Outstanding Returns

Delayed Payments

Tax Information (if applicable)

Bank Transfers

Payment Receipts

Return Forecast

Variance Analysis

Investors can compare projected versus actual performance over time.

3.10 Communication Center

Every investment should include secure communications.

Supported communication types:

Announcements

Questions

Answers

Document Requests

Milestone Updates

Board Announcements (where applicable)

Financial Updates

Compliance Notices

Investor Surveys

System Notifications

Every conversation becomes part of the investment record.

3.11 Document Center

Each investor has a personal document repository.

Documents include:

Investment Agreements

Certificates

Reports

Financial Statements

Distribution Statements

Invoices

Receipts

Legal Notices

Project Updates

Meeting Documents (if shared)

Downloaded Reports

Each document includes version history and download tracking.

3.12 Watchlist

Investors may bookmark opportunities.

Watchlist features:

Price / Funding Updates

Funding Progress

Governance Changes

Compliance Changes

Deadline Reminders

New Documents

AI Alerts

Investors receive notifications when significant changes occur.

3.13 Personalized Recommendations

Masheed should recommend opportunities based on:

Investment history

Preferred sectors

Investment size

Geographic preference

Risk profile

Portfolio diversification

Governance preference

Historical engagement

Recommendations should always explain the reasoning behind the suggestion.

3.14 AI Portfolio Advisor

Instead of a generic chatbot, the AI acts as a portfolio analyst.

Example insights:

> "Your portfolio is heavily concentrated in commercial real estate."
>
> "This new healthcare Waqf opportunity would improve diversification."
>
> "Three investments have improved their governance score this quarter."
>
> "Expected annual returns have increased by 6% compared to last year."
>
> "One project is behind schedule by 45 days."

The AI should never make investment decisions for the investor but should surface relevant information and explain its reasoning.

3.15 Investor Reports

The platform should generate:

Portfolio Statement

Investment Summary

Annual Performance Report

Returns Report

Cash Flow Report

Distribution History

Risk Analysis

Governance Summary

Compliance Summary

Tax Summary (where applicable)

Executive Portfolio Report

Reports should support scheduled delivery and on-demand generation.

3.16 Investor Notifications

Notifications include:

New Opportunity

Funding Progress

Investment Confirmed

Document Available

Return Distributed

Project Update

Governance Update

Compliance Update

Risk Alert

Milestone Completed

Question Answered

Listing Closed

Investors can configure notification preferences by channel and event type.

3.17 Investor Security

Because investors are handling sensitive financial information, the platform should support:

- Multi-factor authentication (MFA).
- Device management.
- Login history.
- Session management.
- Suspicious login detection.
- Download controls for confidential documents.
- Activity history.
- Consent management.

3.18 Portfolio Health Score (Masheed Innovation)

Each investor receives a Portfolio Health Score that summarizes the quality of their investments.

Suggested dimensions:

Dimension	Weight
Diversification	20%
Portfolio Risk	20%
Governance Quality	15%
Compliance Quality	15%
Return Performance	20%
Liquidity & Maturity Profile	10%

This score is informational and helps investors identify opportunities to strengthen their portfolio.

3.19 Institutional Investor Workspace

For institutional investors, provide additional capabilities:

- Multi-user access with permissions.
- Internal approval workflows before investing.
- Investment committee notes.
- Portfolio benchmarking.
- Consolidated reporting.
- Bulk document downloads.
- Custom reporting.
- Relationship manager contact.
- API integration for internal systems (future).

This allows banks, foundations, and family offices to use the platform without relying on individual user accounts.

3.20 Executive Investor Dashboard (Flagship Feature)

The executive dashboard should answer:

How much have I invested?

↓

How is my portfolio performing?

↓

Which projects need attention?

↓

Where are my returns coming from?

↓

Which opportunities match my strategy?

↓

What actions should I take today?

It should combine:

- Live portfolio value.
- Expected distributions.
- Portfolio diversification.
- Governance trends.
- Compliance alerts.
- Project milestones.
- AI-generated priorities.

This dashboard becomes the investor's daily command center.

Major Recommendation — Investor Trust Center

I recommend introducing an **Investor Trust Center**, a unique feature that reinforces confidence in every investment.

Instead of presenting only financial metrics, each opportunity and portfolio should expose measurable trust indicators such as:

- Verified by Masheed ERP.
- Governance Score.
- Compliance Score.
- Last independent valuation date.
- Latest financial reporting date.
- Document completeness.
- Active audit status.
- Number of resolved compliance findings.
- Reporting timeliness.
- Transparency rating.

These indicators create a standardized "trust framework" that investors can compare across opportunities, encouraging Awqaf organizations to improve governance and reporting quality—not just financial performance.

PART 3 — Waqf by Masheed

Chapter 4 — Investment Execution & Capital Management

4.1 Philosophy

Investment is not a payment.

Investment is the beginning of a long-term relationship between:

- Investor
- Waqf
- Asset
- Project
- Beneficiaries
- Governance

Every Riyal entering the platform should always answer:

- Where did it come from?
- Who approved it?
- Which opportunity is it allocated to?
- Where is it today?
- How has it been used?
- What returns has it generated?

Masheed should make this traceability available at every stage.

4.2 Investment Lifecycle

Investor Registration

↓

Opportunity Selection

↓

Investment Simulation

↓

Commitment

↓

Review

↓

Payment

↓

Verification

↓

Allocation

↓

Investment Certificate

↓

Project Execution

↓

Performance Monitoring

↓

Returns Distribution

↓

Investment Closure

Every stage should be timestamped, versioned, and auditable.

4.3 Investment Workflow

An investor should experience the following journey:

Step 1

Review Opportunity

↓

Step 2

Run Investment Calculator

↓

Step 3

Review Expected Returns

↓

Step 4

Accept Terms

↓

Step 5

Identity Confirmation

↓

Step 6

Payment

↓

Step 7

Payment Verification

↓

Step 8

Investment Allocation

↓

Step 9

Digital Investment Certificate

↓

Step 10

Portfolio Update

4.4 Investment Object

Every investment should become its own object.

Investment

Overview

↓

Investor

↓

Opportunity

↓

Financial Details

↓

Documents

↓

Payments

↓

Returns

↓

Updates

↓

Governance

↓

Compliance

↓

Timeline

↓

AI Analysis

4.5 Investment Fields

Identity

Investment ID

Investment Number

Investment Date

Status

Investor

Opportunity

Related Waqf

Related Asset

Investment Manager

Currency

Financial

Investment Amount

Allocated Amount

Committed Amount

Paid Amount

Remaining Amount

Investment Share

Expected ROI

Expected IRR

Expected Payback

Expected Distribution Frequency

Funding Round

Funding Source

Status

Committed

Pending Payment

Paid

Verified

Allocated

Active

Completed

Cancelled

Refunded

Suspended

Documents

Agreement

Certificate

Receipts

Payment Proof

Supporting Documents

Communication History

Audit Log

4.6 Investment Calculator

Before investing, users can simulate scenarios.

Input:

Investment Amount

Holding Period

Expected Revenue

Projected Occupancy

Market Assumptions

Expected Costs

Outputs:

Projected Annual Return

Total Return

Cash Flow

Payback Period

IRR

Sensitivity Analysis

Best Case

Expected Case

Worst Case

The calculator should clearly state that results are projections based on assumptions.

4.7 Payment Management

Supported payment methods should be configurable.

Examples:

- Bank Transfer
- Mada
- Visa
- Mastercard
- Apple Pay
- SADAD (future)
- Open Banking (future)

Each payment records:

Payment ID

Amount

Method

Reference Number

Date

Status

Verification

Gateway Response

Supporting Documents

Refund Status

4.8 Capital Allocation Engine

This is one of Masheed's differentiators.

Money should not simply go into one account.

Instead, allocations should be transparent.

Example:

SAR 1,000,000

↓

Land Development

35%

↓

Construction

40%

↓

Professional Services

8%

↓

Contingency

7%

↓

Reserve

10%

Every allocation should be visible to investors.

4.9 Funding Progress

Every opportunity should display:

Funding Goal

Current Funding

Remaining Amount

Number of Investors

Average Investment

Largest Investment

Funding Velocity

Funding Deadline

Projected Completion

Funding Milestones

4.10 Digital Investment Certificate

Every completed investment generates a digital certificate.

Certificate includes:

Certificate Number

Investor

Opportunity

Waqf

Investment Amount

Date

Reference

QR Code

Verification Link

Digital Signature

Certificate Version

Status

The certificate should be verifiable through a public verification page without exposing confidential information.

4.11 Project Execution Monitoring

Once funding is complete, investors should continue receiving updates.

Project dashboard includes:

Completion Percentage

Current Milestone

Budget Used

Budget Remaining

Schedule Variance

Issues

Photos

Progress Reports

AI Summary

Upcoming Milestones

4.12 Milestone Management

Every project should define milestones.

Each milestone includes:

Milestone Name

Description

Target Date

Actual Date

Budget

Status

Evidence

Photos

Documents

Approvals

AI Progress Summary

Milestone completion can trigger automated investor notifications.

4.13 Returns Distribution

The platform should support configurable distribution schedules.

Examples:

Monthly

Quarterly

Semi-Annual

Annual

Custom

For each distribution:

Distribution ID

Related Investment

Amount

Date

Method

Status

Supporting Statement

Receipt

Variance Explanation

4.14 Post-Investment Reporting

Investors should receive recurring reports including:

Project Progress

Financial Performance

Revenue

Expenses

Budget Variance

Governance Updates

Compliance Updates

Risk Changes

AI Commentary

Reports can be scheduled or generated on demand.

4.15 Exception Management

The platform should detect and manage exceptions such as:

- Delayed project milestones.
- Budget overruns.

- Funding shortages.
- Governance violations.
- Compliance breaches.
- Material legal events.
- Significant changes in projected returns.

Each exception should include:

Severity

Root Cause

Impact

Mitigation Plan

Responsible Party

Expected Resolution

Investor Communication Status

4.16 AI Investment Monitor

The AI should continuously monitor funded opportunities and provide contextual insights.

Examples:

> "Construction progress is 12% behind schedule."
>
> "Current expenditures remain within approved budget."
>
> "Revenue projections have improved due to higher occupancy."
>
> "A recently identified compliance issue may delay the next milestone."

Recommendations should always reference supporting ERP data.

4.17 Investment Timeline

Every investment should maintain a complete historical timeline.

Opportunity Published

↓

Investment Committed

↓

Payment Received

↓

Allocation Completed

↓

Funding Closed

↓

Project Started

↓

Milestone 1

↓

Milestone 2

↓

Revenue Started

↓

Returns Distributed

↓

Investment Closed

Nothing should be removed from the timeline; corrections are appended as new events.

4.18 Administrative Dashboard

Marketplace administrators should monitor:

- Total Capital Raised.
- Active Funding Campaigns.
- Pending Payments.
- Allocation Status.
- Upcoming Distributions.
- Funding Deadlines.
- High-Risk Projects.
- Delayed Milestones.
- Investor Support Requests.

This dashboard becomes the operational control center for marketplace management.

4.19 Reports

The module should generate:

Investment Reports

- Investment Register
- Capital Allocation Report
- Funding Progress Report
- Distribution Report
- Payment Report
- Refund Report
- Project Performance Report

Executive Reports

- Capital Raised by Period
- Investor Participation Analysis
- Funding Success Rate
- Average Investment Size
- Return Performance
- Portfolio Growth

- Marketplace Health Report

4.20 Investment Integrity Engine (Masheed Innovation)

This engine continuously validates that investment capital is being managed according to approved plans.

It compares:

- Planned vs. actual capital deployment.
- Planned vs. actual project milestones.
- Budgeted vs. actual expenditures.
- Governance approvals vs. executed actions.
- Compliance requirements vs. current status.

When significant deviations occur, the engine generates alerts, requests explanations from the organization, and surfaces material issues to marketplace administrators. This creates a transparent chain of accountability from fundraising through project execution.

Major Recommendation — Public Transparency Portal

One feature that I believe could become a defining differentiator for Masheed is a **Public Transparency Portal**.

For each published opportunity, organizations could optionally expose a curated public dashboard showing:

- Funding progress.
- Project milestone completion.
- High-level financial progress.
- Governance status.
- Compliance status.
- Latest updates.
- Photos and videos.

- Community impact metrics.
- Number of beneficiaries served.

This portal would not reveal confidential investor or ERP data, but it would demonstrate accountability and strengthen trust among investors, beneficiaries, regulators, and the public. It aligns closely with the governance and transparency principles underpinning the Awqaf sector and could become one of Masheed's most recognizable features.

PART 4 — Cross-Platform Modules

Chapter 1 — Notifications & Communication Center

1.1 Purpose

The Notification & Communication Center serves as the centralized communication hub across the entire Masheed ecosystem. It ensures that users receive timely, relevant, and actionable information through configurable channels while maintaining a complete communication history for audit and traceability.

The system should support both system-generated events and user-to-user collaboration, ensuring that no important governance, compliance, investment, or operational event is missed.

1.2 Communication Architecture

Communication Center

├── Notifications
├── Inbox
├── Announcements
├── Chat
├── Email
├── SMS
├── Push Notifications

├── WhatsApp (Future)
├── In-App Alerts
├── Scheduled Messages
├── Broadcast Messages
└── Communication Logs

1.3 Notification Categories

The platform should support categorized notifications.

ERP

- Asset Updates
- Financial Alerts
- Compliance Alerts
- Governance Actions
- Maintenance Requests
- Consultations
- Workflow Approvals
- Risk Alerts
- AI Recommendations

Marketplace

- Listing Status
- New Investments
- Funding Progress
- Investor Questions
- Returns Distribution
- Milestone Updates
- Listing Approval
- Opportunity Expiration

System

- Login Alerts
- Password Changes
- Role Changes

- Security Alerts
- New Features
- Scheduled Maintenance

1.4 Notification Object

Each notification should include:

- Notification ID
- Category
- Title
- Description
- Related Module
- Related Record
- Priority
- Severity
- Sender
- Recipient(s)
- Delivery Channel
- Created Date
- Read Date
- Expiration Date
- Status
- Action Button
- Deep Link

1.5 Priority Levels

Notifications should support configurable priorities.

- Critical
- High
- Medium
- Low
- Informational

Critical notifications may require acknowledgment before dismissal.

1.6 Delivery Channels

Users should be able to receive notifications through multiple channels.

Supported channels:

- In-App
- Email
- SMS
- Push Notifications
- Browser Notifications
- WhatsApp (Future)
- Microsoft Teams (Future)
- Slack (Future)

Organizations can enable or disable channels globally.

1.7 Notification Preferences

Each user can configure:

- Preferred channels
- Quiet hours
- Language
- Daily digest
- Weekly summary
- Immediate alerts
- AI recommendation frequency
- Security alerts (mandatory)

Preferences can also be overridden by organizational policies for critical events.

1.8 Notification Center

The notification center should support:

- Unread counter
- Search
- Filters
- Categories
- Pinning
- Snooze
- Archive
- Bulk actions
- Read all
- Mark as important

1.9 Inbox

The inbox stores all communication.

Supported message types:

- Direct Messages
- Workflow Requests
- Approval Requests
- Consultation Messages
- Marketplace Messages
- AI Messages
- System Messages

Messages should support attachments, mentions, reactions, and threaded replies.

1.10 Chat

Instead of a generic chat, conversations should be contextual.

Examples:

Every asset can have a discussion.

Every investment can have a discussion.

Every consultation can have a discussion.

Every workflow can have a discussion.

Every listing can have a discussion.

Supported features:

- Threaded conversations
- Mentions (@User)
- File sharing
- Voice notes (future)
- Read receipts
- Typing indicators
- Emoji reactions
- AI conversation summary

1.11 Announcements

Organizations can publish announcements.

Examples:

- New Policy
- Board Decision
- System Update
- Regulatory Announcement
- Upcoming Event
- Marketplace Campaign

Each announcement includes:

- Audience
- Start Date
- End Date
- Priority
- Attachments
- Required Acknowledgment
- Read Statistics

1.12 Workflow Notifications

Every workflow automatically generates notifications.

Example:

Request Submitted

↓

Assigned

↓

Under Review

↓

Approved

↓

Completed

↓

Closed

Users receive updates at each stage.

1.13 AI Notification Assistant

The AI should reduce notification fatigue.

Instead of sending dozens of alerts, it can group them intelligently.

Example:

Instead of:

- 12 document expirations
- 8 maintenance reminders
- 5 policy reviews

The AI sends:

> "Today requires your attention in three areas: document renewals, maintenance planning, and policy reviews. There are 25 pending actions affecting five Waqf entities."

The user can then expand each section.

1.14 Escalation Rules

Notifications should escalate automatically.

Example:

Approval pending 48 hours

↓

Reminder

↓

Still pending

↓

Manager notified

↓

Still pending

↓

Department Head

↓

Still pending

↓

Executive Dashboard

Escalation rules should be configurable.

1.15 Communication Logs

Every communication event should be recorded.

Fields:

- Communication ID
- Sender
- Recipient
- Channel
- Related Record
- Timestamp
- Delivery Status
- Read Status
- Response Time
- Attachments
- Audit Reference

1.16 Broadcast Center

Administrators can send broadcast messages.

Audience examples:

- All Users
- Finance Team
- Investors
- Board Members
- Marketplace Reviewers
- Consultants
- Organization-specific Users

Scheduling and targeting rules should be supported.

1.17 Notification Templates

Templates should be configurable using placeholders.

Example:

Hello {{UserName}}

Your investment in {{OpportunityName}} has been successfully confirmed.

Amount:
{{InvestmentAmount}}

Thank you.

Supported placeholders include user, organization, Waqf, asset, workflow, and investment variables.

1.18 Reports

The module should provide:

- Notification Delivery Report
- Read Rate Analysis
- Communication Activity
- Response Time Analysis
- Announcement Engagement
- Escalation Report
- Broadcast Performance

1.19 Command Center Widget

The executive dashboard should summarize communications:

- Critical Notifications
- Pending Approvals
- Unread Messages

- Escalated Issues
- AI Daily Briefing
- Upcoming Deadlines

Executives should immediately know which communications require attention without opening multiple modules.

Major Recommendation — Unified Activity Feed

Instead of forcing users to open different modules, Masheed should provide a unified activity feed similar to GitHub, Notion, or Microsoft Teams.

Example:

09:05

Investment Opportunity approved

↓

09:12

Board Decision recorded

↓

09:18

Maintenance request completed

↓

09:35

Compliance score increased

↓

09:40

AI Recommendation generated

↓

09:42

Investor committed SAR 250,000

↓

09:55

Project milestone completed

Every item should be clickable, filterable, and linked directly to its source record. This feed becomes the "heartbeat" of the platform, giving users real-time visibility into everything happening across the ERP and the marketplace.

PART 4 — Cross-Platform Modules

Chapter 2 — User, Roles & Authorization Management

2.1 Purpose

The Authorization Management module provides enterprise-grade identity, access control, and security for the entire Masheed ecosystem.

It governs access across:

- Masheed ERP
- Waqf by Masheed
- Administration Portal
- Mobile Applications
- APIs
- AI Services

- Future Integrations

The system should ensure every user can access **only the information, actions, and workflows they are explicitly authorized to perform**, while maintaining complete traceability of all permission changes.

2.2 Identity Architecture

Identity Platform

├── Organizations
├── Users
├── Roles
├── Permission Sets
├── Groups
├── Teams
├── Delegation
├── API Clients
├── Authentication
├── Authorization
├── Sessions
├── MFA
├── SSO
└── Audit

2.3 User Types

Internal Users

- Super Administrator
- Organization Administrator
- CEO
- Nazer
- CFO
- Finance Manager
- Asset Manager
- Investment Manager
- Governance Officer
- Compliance Officer

- Legal Officer
- Risk Manager
- Procurement Officer
- HR Officer
- Internal Auditor
- Board Member
- Committee Member
- Executive Assistant

Marketplace Users

- Investor
- Institutional Investor
- Opportunity Owner
- Marketplace Reviewer
- Marketplace Administrator
- Support Team

External Users

- Consultant
- Auditor
- Valuation Company
- Law Firm
- Engineering Consultant
- Government Reviewer
- External Accountant

System Users

- AI Agents
- API Integrations
- Scheduled Jobs
- Webhooks
- Background Workers

2.4 User Profile

Each user should contain:

Identity

- User ID
- Employee ID (optional)
- Arabic Name
- English Name
- Email
- Mobile
- Username
- Profile Photo
- Organization
- Department
- Position
- Manager
- Preferred Language
- Time Zone

Security

- MFA Enabled
- Password Status
- Last Login
- Last Password Change
- Failed Login Attempts
- Active Sessions
- Trusted Devices
- IP Restrictions
- Account Status

Authorization

- Assigned Roles
- Permission Sets
- Teams

- Groups
- Delegated Permissions
- Temporary Permissions
- Data Scope
- Approval Limits

2.5 Role Hierarchy

Platform Super Admin

↓

Platform Operations

↓

Organization Admin

↓

Executive Management

↓

Department Managers

↓

Specialists

↓

Employees

↓

External Users

↓

Read Only

Organizations can customize this hierarchy.

2.6 Permission Model

Permissions should be granular.

Examples:

Module Permissions

- View Assets
- Create Assets
- Edit Assets
- Delete Assets
- Archive Assets
- Export Assets

Financial

- Create Payments
- Approve Payments
- Reverse Journals
- Close Accounting Period
- Export Financial Reports

Marketplace

- Create Listing
- Review Listing
- Approve Listing
- Publish Listing
- Suspend Listing
- View Investors

Governance

- Create Board Meeting
- Vote
- Approve Decisions
- Manage Policies
- Manage Risks

2.7 Permission Levels

Every object should support:

View

Create

Edit

Delete

Approve

Reject

Assign

Export

Print

Share

Archive

Restore

Configure

Administrate

Execute Workflow

Override

2.8 Record-Level Security

Users may only access records matching defined rules.

Examples:

Asset belongs to Organization A

↓

Only Organization A

Finance

Asset Manager

CEO

can access it.

Another example:

Investor

↓

Can only access

their own investments

their own documents

their own reports

2.9 Field-Level Security

Some users may open a record but not every field.

Example

Asset

Market Value

CEO ✓

Finance ✓

Asset Manager ✓

Maintenance ✗

Investor ✗

Another example

Salary

HR ✓

CEO ✓

Manager ✗

2.10 Attribute-Based Security (ABAC)

Access decisions can consider attributes.

Examples:

User Department

Location

Organization

Country

Role

Project

Investment Amount

Risk Level

Workflow Stage

Classification

Time

Device

IP Address

Example Rule

Finance Managers

can approve expenses

ONLY

Department = Finance

AND

Amount < SAR 100,000

AND

Working Hours

AND

Within Assigned Organization

2.11 Temporary Permissions

Managers can grant temporary access.

Fields

Who Granted

Who Received

Reason

Start Date

End Date

Approval

Automatic Expiration

Audit Log Reference

2.12 Delegation

Example

CEO on vacation

↓

Delegates

Financial Approvals

↓

CFO

↓

Valid

July 1–15

↓

Automatically expires

Delegation applies only to selected permissions.

2.13 Groups & Teams

Support:

Departments

Projects

Investment Committees

Board Committees

Regional Offices

Working Groups

Review Teams

Notifications, workflows, and approvals can target groups rather than individual users.

2.14 Authentication

Supported methods:

- Email & Password
- MFA
- SMS OTP
- Authenticator Apps
- Passkeys (future)
- Biometric Login (mobile)
- SAML SSO
- OAuth 2.0
- OpenID Connect
- Government Identity Providers (future)

2.15 Session Management

Administrators can view:

- Active Sessions
- Device
- Browser
- Operating System
- IP Address
- Login Time
- Last Activity
- Risk Score

Actions:

- Terminate Session
- Force Logout
- Reset MFA
- Require Password Change

2.16 AI Permission Advisor

The AI analyzes permission assignments.

Examples:

> "12 users have excessive financial permissions."
>
> "Three users have not used their approval permissions in six months."
>
> "A temporary permission expired but remains active due to configuration."
>
> "The Compliance Officer lacks access to the new Risk module."

2.17 Reports

- User Register
- Role Matrix
- Permission Matrix
- Login Activity
- Failed Logins
- Delegation Report
- Temporary Access Report

- MFA Adoption
- Access Review Report
- Segregation of Duties Report

2.18 Enterprise Recommendation

Implement **Segregation of Duties (SoD)** rules from the beginning.

Examples:

A user **cannot** simultaneously:

- Create a payment and approve the same payment.
- Create a listing and perform the final listing approval.
- Record an audit finding and close the same finding.
- Create a governance policy and approve its publication without an independent reviewer.

This dramatically strengthens governance and is expected by enterprise customers.

MASHEED Master Product Specification

PART 4 — Cross-Platform Modules

Chapter 3 — Audit Logs & Activity History

3.1 Purpose

Every meaningful action in Masheed must be traceable.

The audit system should be immutable, searchable, exportable, and compliant with enterprise governance requirements.

Nothing important should happen without leaving an audit trail.

3.2 What Must Be Audited

The system should log:

- Logins and logouts.
- Failed authentication attempts.
- Password changes.
- MFA changes.
- User creation and deactivation.
- Permission changes.
- Role assignments.
- Record creation.
- Record updates.
- Record deletion requests.
- Soft deletes and restores.
- Workflow approvals.
- Workflow rejections.
- Financial postings.
- Investment transactions.
- AI-generated recommendations (and whether they were accepted or dismissed).
- File uploads/downloads.
- API requests.
- Configuration changes.
- Report exports.
- Bulk operations.

3.3 Audit Record

Every audit event includes:

- Audit ID
- Timestamp (UTC)
- User / Service Account
- Organization
- Module
- Entity Type

- Entity ID
- Action
- Before Values
- After Values
- Changed Fields
- IP Address
- Device
- Browser
- Session ID
- Correlation ID (to trace multi-step workflows)
- Source (Web, Mobile, API, Integration)
- Outcome (Success / Failure)
- Reason (where required)

3.4 Version History

Every business record should support versioning.

Users can:

- View all previous versions.
- Compare versions side-by-side.
- Restore an earlier version (where business rules permit).
- See who made each change and why.

3.5 Timeline View

Every record should expose a unified timeline.

Example for an Asset:

Created

↓

Valuation Updated

↓

Maintenance Scheduled

↓

Insurance Renewed

↓

Listed for Investment

↓

Funding Completed

↓

Compliance Review

↓

Inspection Completed

This timeline becomes the historical narrative of the record.

3.6 Search & Investigation

Audit logs should support advanced filters:

- Date range.
- User.
- Organization.
- Module.
- Entity.
- Action.
- IP address.
- Device.
- Outcome.
- Severity.

Investigators should be able to reconstruct an entire workflow from a single Correlation ID.

3.7 Retention & Integrity

- Configurable retention policies.
- Tamper-evident storage.
- Export for regulators.
- Legal hold capability.
- Cryptographic integrity checks (future enhancement).

PART 4 — Cross-Platform Modules

Chapter 4 — Settings & Configuration

4.1 Philosophy

Masheed should be **configured, not customized**.

Nearly every business rule should be manageable through configuration, allowing organizations to adapt the platform without requiring software development.

4.2 Configuration Hierarchy

Platform

↓

Tenant (Organization)

↓

Waqf

↓

Department

↓

Module

↓

Workflow

↓

User

Higher-level settings define defaults, while lower levels may override them where permitted.

4.3 Platform Settings

Examples:

- Supported Languages
- Supported Currencies
- Time Zones
- Number Formats
- Date Formats
- Email Providers
- SMS Providers
- Payment Providers
- File Storage
- AI Providers
- Security Policies
- Branding Defaults
- Maintenance Mode

4.4 Organization Settings

Each organization can configure:

- Company profile.
- Logo and branding.
- Contact details.
- Working days.
- Business hours.
- Fiscal year.
- Approval thresholds.
- Notification defaults.
- Default document templates.
- Default workflows.
- Department structure.
- Cost centers.
- Organizational hierarchy.

4.5 Module Settings

Every module should expose configurable options.

Examples:

Assets:

- Custom asset categories.
- Valuation frequency.
- Maintenance intervals.

Finance:

- Fiscal calendar.
- Chart of accounts.
- Approval limits.
- Budget rules.

Marketplace:

- Listing review stages.
- Required documents.
- Funding limits.
- Visibility rules.

Governance:

- Board term limits.
- Committee structures.
- Policy review cycles.

4.6 Dynamic Fields

Administrators should be able to add custom fields to almost any entity.

Supported types:

- Text
- Number
- Currency
- Date
- Date & Time
- Boolean
- Dropdown
- Multi-select
- User
- Organization
- File
- URL
- Formula
- Lookup
- Rich Text

Fields should support validation, conditional visibility, and API exposure.

4.7 Branding

Organizations can configure:

- Logos.
- Color palette.
- Email templates.
- Report templates.

- PDF headers and footers.
- Public marketplace branding (within platform guidelines).

4.8 Localization

Support:

- Arabic (RTL)
- English (LTR)

Future-ready architecture should allow adding additional languages without code changes.

Localization should include:

- UI text.
- Dates.
- Numbers.
- Currency.
- Reports.
- Email templates.
- Notifications.

4.9 Feature Flags

Administrators should enable or disable features at:

- Platform level.
- Organization level.
- Beta group.
- Individual user (for testing).

This supports phased rollouts and enterprise pilots.

4.10 Configuration Packages

Organizations should be able to export and import configuration packages containing:

- Roles.
- Permission sets.
- Workflows.
- Notification templates.
- Document templates.
- Custom fields.
- Approval matrices.
- Settings.

This makes onboarding new organizations dramatically faster.

PART 4 — Cross-Platform Modules

Chapter 5 — Workflow & Business Process Engine (WBPE)

5.1 Vision

The Workflow & Business Process Engine (WBPE) is the orchestration layer that powers every business process across Masheed ERP and Waqf by Masheed.

Rather than embedding workflows directly into each module, the engine provides a centralized, configurable platform that allows administrators to design, automate, monitor, and optimize business processes without software development.

Every approval, review, task, escalation, notification, and AI checkpoint should be orchestrated by this engine.

5.2 Workflow Philosophy

Everything is a workflow.

Examples:

New Waqf

↓

Register

↓

Review

↓

Approve

↓

Activate
Investment Listing

↓

Documents

↓

Legal

↓

Finance

↓

Governance

↓

Compliance

↓

Investment Committee

↓

Publish
Maintenance Request

↓

Submit

↓

Review

↓

Assign

↓

Execute

↓

Inspect

↓

Close
Board Decision

↓

Meeting

↓

Vote

↓

Approve

↓

Assign Tasks

↓

Monitor

↓

Close

Nothing should bypass the Workflow Engine.

5.3 Engine Architecture

Workflow Engine

├── Workflow Designer
├── Process Templates
├── Stage Designer
├── Approval Engine
├── Assignment Engine
├── Rule Engine
├── Automation Engine
├── SLA Engine
├── Notification Engine
├── AI Decision Engine
├── Escalation Engine
├── Forms Engine
├── Document Engine
├── Audit Engine
├── Analytics Engine
└── API Engine

5.4 Workflow Object

Every workflow becomes a first-class object.

Fields

- Workflow ID
- Workflow Name
- Description
- Category
- Module
- Organization
- Status
- Version
- Owner
- Created Date
- Last Published
- Active Version
- Draft Version

5.5 Workflow Categories

ERP

- Asset Management
- Finance
- Governance
- Compliance
- Consultation
- Procurement
- Maintenance
- Risk
- Audit

Marketplace

- Listing Review
- Investment Approval
- Due Diligence
- Investor Verification
- Distribution Approval

Platform

- User Provisioning
- Organization Onboarding
- API Access
- Subscription Changes

- Feature Requests

5.6 Workflow Designer

Administrators should build workflows visually.

Start

↓

Create Record

↓

Assign Reviewer

↓

Wait

↓

Approval

↓

AI Review

↓

Decision

↓

Notification

↓

Finish

Every node should be draggable.

No code required.

5.7 Stage Object

Each workflow contains stages.

Fields

Stage ID

Stage Name

Description

Owner

SLA

Status

Entry Conditions

Exit Conditions

Required Permissions

Required Fields

Required Documents

Notifications

Automation

Escalation

AI Actions

5.8 Entry Requirements

Every stage may define:

Required Fields

Example

Asset Registration

Required

✓ Asset Name

✓ Waqf

✓ Asset Type

✓ GPS

✓ Ownership

Cannot continue until completed.

Required Documents

Example

Listing Review

✓ Valuation Report

✓ Title Deed

✓ Financial Statements

✓ Board Resolution

Required Approvals

Example

Cannot enter Investment Committee stage

until

Legal ✓

Finance ✓

Governance ✓

Compliance ✓

5.9 Exit Rules

A stage finishes only when rules are satisfied.

Example

Finance Review

Must satisfy:

Budget verified

Financial statements uploaded

ROI calculated

Reviewer signed

No critical findings

↓

Move to next stage

5.10 Assignment Engine

Assignments support:

Specific User

Role

Department

Committee

Dynamic Rule

Round Robin

Least Busy User

Geographic Assignment

Organization

AI Recommendation

Example

Engineering Report

↓

Assign

Engineering Consultant

↓

If unavailable

↓

Next Certified Consultant

5.11 Approval Engine

Supported approvals

Single Approval

Sequential Approval

Parallel Approval

Majority Vote

Unanimous Vote

Committee Vote

Conditional Approval

Delegated Approval

Weighted Voting (future)

Example

CEO

↓

Finance

↓

Legal

↓

Board

↓

Approved

or

Finance

Legal

Governance

(all parallel)

↓

Investment Committee

5.12 Conditional Logic

Examples

If

Investment > SAR 5M

↓

Add Board Review

If

Risk High

↓

Require Compliance Review

If

Government Asset

↓

Require Government Approval

If

International Investor

↓

Require Additional Verification

Rules support:

AND

OR

NOT

Nested conditions

Expressions

Calculated fields

External API responses

5.13 Forms Engine

Every stage may have its own form.

Supported components

Text

Rich Text

Currency

Number

Dropdown

Multi Select

Checkbox

Radio

User

Organization

Asset Lookup

Investment Lookup

Date

Time

File Upload

Repeating Tables

Digital Signature

AI Generated Fields

Formula Fields

Hidden Fields

Conditional Fields

5.14 Automation Engine

Automations include

Create Task

Create Meeting

Generate Document

Send Notification

Assign User

Generate AI Summary

Create Consultation

Update Status

Call API

Generate Report

Schedule Reminder

Start Another Workflow

Example

Investment Approved

↓

Generate Agreement

↓

Generate Certificate

↓

Notify Investor

↓

Update Portfolio

↓

Sync ERP

5.15 SLA Engine

Every stage supports SLAs.

Example

Legal Review

Target

5 Days

↓

After 3 Days

Reminder

↓

After 5 Days

Escalate

↓

After 7 Days

Manager

↓

After 10 Days

Executive Dashboard

Dashboard

SLA Met

SLA Breached

Average Completion

Longest Running

5.16 Escalation Engine

Escalation actions

Reminder

Manager Notification

Reassignment

Priority Increase

Executive Alert

Temporary Delegation

Workflow Pause

Workflow Cancellation

5.17 AI Workflow Assistant

AI assists throughout the workflow.

Examples

"Required document appears expired."

"Previous similar workflow averaged 4 days."

"Recommend assigning Reviewer Ahmed based on workload."

"Financial assumptions differ significantly from historical projects."

"Predicted approval probability: 92%."

"Expected completion in 11 days."

AI suggestions remain advisory unless explicitly configured otherwise.

5.18 Workflow Analytics

Every workflow should provide analytics.

KPIs

Average Completion Time

Average Time Per Stage

Approval Rate

Rejection Rate

Rework Rate

SLA Compliance

Bottlenecks

Most Delayed Stage

Reviewer Performance

AI Prediction Accuracy

Automation Savings

Visualizations

Funnel

Cycle Time

Stage Heatmap

Completion Trend

Organization Comparison

5.19 Workflow Templates

Masheed should ship with ready-made templates.

Examples

ERP

- Asset Registration
- Asset Disposal
- Asset Valuation
- Maintenance

- Procurement
- Budget Approval
- Payment Approval
- Beneficiary Distribution
- Board Decision
- Internal Audit
- Consultation Request

Marketplace

- Listing Application
- Listing Review
- Due Diligence
- Investor Verification
- Investment Subscription
- Capital Distribution
- Project Monitoring

Administration

- Organization Onboarding
- User Onboarding
- Permission Request
- Feature Activation
- Subscription Upgrade

5.20 Business Rules Engine

Separate business rules from workflows.

Example

Rule

Maximum approval amount

↓

SAR 100,000

↓

Used by

Finance

Procurement

Investments

Marketplace

Changing the rule updates all dependent workflows automatically.

5.21 Workflow Versioning

Every workflow supports:

Draft

Published

Archived

Rollback

Comparison

Change History

Approval before publishing

Organizations can test a draft workflow before activating it.

5.22 Cross-Workflow Orchestration

One workflow can trigger another.

Example

Asset Acquisition

↓

Generate Legal Review

↓

Generate Insurance Workflow

↓

Generate Initial Valuation

↓

Generate Compliance Review

↓

Generate Maintenance Plan

↓

Ready

This enables end-to-end automation without manual intervention.

5.23 Workflow APIs

Every workflow should expose APIs for:

- Start Workflow
- Advance Stage
- Reject Stage
- Pause Workflow
- Resume Workflow
- Cancel Workflow
- Query Status
- Upload Documents
- Retrieve Timeline
- Subscribe to Events

This allows integrations with government systems, payment providers, CRMs, or future mobile applications.

5.24 Workflow Command Center (Flagship Feature)

This should become one of the most powerful screens in Masheed.

Executives see:

Total Active Workflows

↓

Pending Approvals

↓

SLA Breaches

↓

Critical Escalations

↓

AI Predicted Delays

↓

Average Completion Time

↓

Bottleneck Departments

↓

Organization Performance

↓

Today's Priority Actions

Users can drill down from every KPI directly into the affected workflows.

5.25 Process Mining & Continuous Improvement (Future Enterprise Edition)

This is a feature very few platforms in the region offer.

The engine should continuously analyze completed workflows to discover:

- Repeated bottlenecks.
- Unnecessary approval steps.
- Frequently returned submissions.
- High-performing reviewers.
- Average processing times by organization.
- AI recommendations for workflow optimization.

For example:

> "The Legal Review stage causes 42% of total processing delays."
>
> "Removing one duplicate approval step could reduce listing approval time by 18%."

These insights help organizations improve operations without manually analyzing reports.

Major Architectural Recommendation — Event-Driven Platform

This is probably my strongest technical recommendation for the entire Masheed architecture.

Don't make modules call each other directly.

Instead, introduce an **Event Bus**.

For example:

Investment Approved

↓

Event Published

↓

Finance Module reacts

↓

Notification Module reacts

↓

Document Module reacts

↓

Portfolio Module reacts

↓

Audit Module reacts

↓

AI Module reacts

↓

Marketplace updates

Every major action should publish events, and every module subscribes only to the events it needs.

This gives you:

- Loose coupling between modules.
- Easier scalability.
- Better maintainability.
- Simpler integrations.
- Future microservices readiness.
- Real-time updates across ERP and Marketplace.

This architecture will make Masheed significantly easier to evolve over the next 5–10 years.

PART 5 — Platform Administration Portal

1.1 Vision

The Platform Administration Portal is the centralized management system for the Masheed SaaS platform.

It provides complete visibility and control over all organizations, users, subscriptions, platform services, AI resources, integrations, monitoring, security, billing, support, and marketplace operations.

It is isolated from customer environments and follows strict security and segregation-of-duty principles.

1.2 Portal Architecture

Platform Administration

├── Executive Dashboard
├── Organization Management
├── Subscription & Licensing
├── Marketplace Operations
├── User Administration
├── Support Center
├── AI Operations
├── Platform Configuration
├── Feature Flags
├── Monitoring & Observability
├── Security Operations
├── Audit Center
├── Integrations
├── Billing
├── Incident Management
├── System Jobs
├── Notifications
├── Reports
└── Developer Center

1.3 Platform Roles

The portal should support separate platform roles.

Executive

- Platform Owner
- CEO
- CTO
- COO

Operations

- Platform Administrator
- Tenant Manager
- Marketplace Operations Manager
- Customer Success Manager
- Support Manager

Security

- Security Administrator
- Compliance Officer
- SOC Analyst
- Identity Administrator

Technical

- DevOps Engineer
- Cloud Engineer
- AI Engineer
- Database Administrator
- Integration Engineer

Financial

- Billing Administrator
- Finance Manager

Read Only

- Executive Viewer
- Auditor

1.4 Executive Dashboard

This becomes the heartbeat of the business.

Widgets include:

Platform Health

Total Organizations

Active Users

Monthly Active Users

Marketplace Capital Raised

Number of Active Investments

Platform Revenue

MRR

ARR

Churn Rate

Trial Conversions

Support Tickets

Critical Incidents

System Availability

Average API Response Time

AI Usage

Storage Consumption

Background Jobs

Security Alerts

Failed Logins

Top Organizations

Growth Trends

1.5 Organization Management

Every customer organization becomes a tenant.

Organization Profile

- Organization ID
- Organization Name
- Arabic Name
- Registration Number
- Subscription Plan
- Status
- Region
- Industry
- Onboarding Date
- Renewal Date
- Assigned Customer Success Manager

Organization Actions

Create

Suspend

Archive

Delete (Soft)

Reactivate

Transfer Ownership

Clone Configuration

Upgrade Plan

Downgrade Plan

Reset Trial

Generate Health Report

1.6 Organization Health Score

Each organization receives a health score based on:

- User adoption.
- Workflow completion.
- AI utilization.
- Login frequency.
- Marketplace participation.
- Data completeness.
- Configuration maturity.
- Support ticket volume.
- Compliance posture.

Organizations with declining health are automatically flagged for Customer Success follow-up.

1.7 Subscription & Licensing

Supported plans

- Trial

- Starter
- Professional
- Enterprise
- Government
- Custom

Each subscription tracks:

- Plan
- Billing Cycle
- Renewal Date
- License Count
- Active Users
- Storage Usage
- AI Credits (if applicable)
- Marketplace Access
- Enabled Modules
- Add-ons

1.8 Marketplace Operations

Internal team dashboard.

Manage:

- Listing Approvals
- Marketplace Moderation
- Suspended Listings
- Fraud Review
- Investor Complaints
- Organization Verification
- Featured Opportunities
- Public Content Approval

KPIs:

- Listings Awaiting Review
- Average Review Time
- Funding Volume
- Investor Growth
- Organization Growth

1.9 User Administration

Platform admins can:

Search users globally.

Reset passwords.

Unlock accounts.

Reset MFA.

Transfer ownership.

Deactivate users.

View login history.

Terminate sessions.

View organization memberships.

Assign support access.

No admin should directly impersonate a user without explicit logging and authorization.

1.10 Support Center

Every support request becomes a structured case.

Fields:

- Case ID
- Customer
- Priority
- Severity
- Module
- Assigned Engineer
- SLA

- Status
- Root Cause
- Resolution
- Linked Audit Events
- Linked Errors

Support agents should be able to request temporary diagnostic access through an approval workflow.

1.11 AI Operations

Manage:

- AI Providers
- Active Models
- Prompt Templates
- AI Agents
- Rate Limits
- Token Usage
- AI Costs
- AI Performance
- AI Error Logs
- AI Accuracy Metrics

The portal should allow switching AI providers without changing business modules.

1.12 Feature Flags

Roll out features gradually.

Scopes:

- Global
- Subscription Plan
- Organization
- Region
- Beta Group
- Individual Tenant

Examples:

Enable Tokenization Module

Enable Mobile App

Enable AI Consultation

Enable New Marketplace UI

Enable Government Integration

Enable Arabic OCR

1.13 Monitoring & Observability

Platform health should include:

Application Health

Database Health

Queue Health

Background Jobs

Cache Health

API Health

Storage

Search Index

AI Services

Email Provider

SMS Provider

Payment Gateway

Government Integrations

Map Services

Every service reports:

Status

Latency

Availability

Error Rate

Last Incident

1.14 Security Operations Center (SOC)

Monitor:

Failed Logins

Impossible Travel

Brute Force Attempts

Suspicious API Calls

Privilege Escalation

Inactive Admin Accounts

Expired Certificates

Compromised Credentials

Unusual Data Exports

Abnormal AI Usage

Security alerts should integrate with SIEM platforms in enterprise deployments.

1.15 Billing Center

Manage:

Invoices

Payments

Credits

Discounts

Coupons

Taxes

Refunds

Usage Billing

Marketplace Fees

Enterprise Contracts

Billing Reports

Future support:

Consumption-based billing for AI usage and storage.

1.16 Incident Management

Every outage or operational issue should be tracked.

Fields:

- Incident ID
- Severity
- Impacted Services
- Start Time
- Resolution Time
- Root Cause

- Timeline
- Customer Impact
- Postmortem
- Preventive Actions

Executives should see live incident status from the dashboard.

1.17 Background Jobs

Monitor:

Scheduled Jobs

Failed Jobs

Retries

Long-running Jobs

Queue Length

Average Processing Time

Dead Letter Queue

Examples:

Nightly Reports

AI Analysis

Marketplace Synchronization

Notification Delivery

Document OCR

Backup Jobs

1.18 Global Audit Center

Unlike tenant audit logs, this tracks platform operations.

Examples:

Subscription upgraded.

Tenant suspended.

Feature flag changed.

Admin login.

Support access granted.

Marketplace listing removed.

AI provider switched.

Configuration changed.

These logs should never be editable.

1.19 Reports

Executive Reports

- Platform Growth
- Revenue
- Customer Acquisition
- Churn
- AI Usage
- Marketplace Performance
- Customer Health
- Incident Analysis
- Security Overview

Operational Reports

- Support SLA
- Tenant Adoption
- Login Statistics
- API Usage
- Storage Growth

- Queue Performance
- Billing Status
- Marketplace Moderation

1.20 Developer Center

Manage:

API Keys

OAuth Clients

Webhooks

SDK Access

Sandbox Environments

API Usage Analytics

Rate Limits

Developer Documentation

Webhook Logs

Version Management

Future:

Partner Marketplace

1.21 Disaster Recovery & Business Continuity

The administration portal should include operational controls for resilience:

- Backup management.
- Restore testing.

- Disaster recovery plans.
- Multi-region failover status.
- Recovery Point Objective (RPO) monitoring.
- Recovery Time Objective (RTO) monitoring.
- Business continuity checklists.
- Scheduled resilience drills.
- Backup verification reports.

Platform administrators should be able to verify that backups are valid and periodically test restoration procedures without affecting production tenants.

1.22 Customer Success Workspace

This is a module that many SaaS platforms overlook but becomes invaluable as the customer base grows.

Each organization should have a dedicated Customer Success workspace containing:

- Onboarding progress.
- Assigned Customer Success Manager.
- Adoption score.
- Training completion.
- Feature utilization.
- Open support cases.
- Executive contacts.
- Success plans.
- Renewal risk.
- Expansion opportunities.
- Meeting history.
- Health trends.

The system can proactively recommend customer engagement when adoption declines or renewal risk increases.

1.23 Marketplace Governance Center

Since Masheed operates a regulated investment marketplace, platform administrators need a dedicated governance console to oversee marketplace integrity.

Capabilities include:

- Monitor listing review queues.
- Track review SLAs.
- Review flagged opportunities.
- Manage public content.
- Handle investor complaints.
- Suspend or reinstate listings.
- Review AI-generated risk alerts.
- Escalate potential fraud or governance issues.
- Publish marketplace-wide announcements.
- Monitor transparency and reporting compliance.

This module ensures the marketplace remains trusted, fair, and compliant while providing administrators with centralized operational oversight.

Final Recommendation — Platform Command Center

Rather than relying on multiple disconnected dashboards, I recommend creating a **Platform Command Center** as the first screen every Masheed operator sees after logging in.

It should consolidate:

- Platform health.
- Critical incidents.
- Security alerts.
- Customer health.
- Marketplace activity.
- AI operations.
- Support queues.
- Financial metrics.
- Infrastructure status.
- Pending approvals.
- Executive KPIs.

From this single screen, administrators should be able to drill into any area of the platform with one click. This becomes the operational nerve center for Masheed and gives leadership, operations, engineering, and customer success a unified view of the entire ecosystem.